

DATA DRIVEN PREDICTON OF STUDENT MENTAL IILNESS AND DEPRESSION INDICATORS USING MACHINE LEARNING

PROJECT

Submitted in partial fulfilment
of the requirements for the degree of

B.Sc Honours (Data Science)

by

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CERTIFICATE

This is to certify that the PROJECT WORK entitled

"DATA DRIVEN PREDICTION OF STUDENT MENTAL ILLNESS AND DEPRESSION INDICATORS USING MACHINE LEARNING"

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DECLARATION

I hereby declare that the project work presented, **“DATA DRIVEN PREDICTION OF STUDENT MENTAL ILLNESS AND DEPRESSION INDICATORS USING MACHINE LEARNING”** is original and has been carried out by myself under the supervision of Mr. SHREESH VITHAL DESHPANDE, Department of Mathematics and Statistics, Bhavan’s Vivekananda College of Science, Humanities and Commerce, Hyderabad, India. All prior work upon which this project is built is cited at appropriate places. I further declare that this work has not been submitted in part or full for the award of any degree of this or any other university.

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To

**My Beloved
Family**

DR. P.R. VENUGOPAL

MS. B. SWAPNA

&

MS. P.V. NAGA DEEPIKA

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Abstract

DATA DRIVEN PREDICTION OF STUDENT MENTAL ILLNESS AND DEPRESSION INDICATORS USING MACHINE LEARNING

Student mental health has turned into a rising issue over the past few years, primarily because of the increasing academic pressures, lifestyle, and social issues. In the present project, the focus of the research work is the prediction of student depression using machine learning techniques, keeping various influencing factors into consideration, such as academic pressures, work pressures, CGPA, study satisfaction, job satisfaction, and daily work/study hours. The data set used for the purpose of the project has been obtained from Kaggle, which has multiple attributes, thus aiding the process of pattern identification.

The prime aim of the present study is the development of an efficient system, which can classify whether the student suffers from depression or not. Pre-processing techniques have been used on the data set, including data cleaning, encoding, and normalization. Multiple machine learning algorithms including Logistic Regression, K-Nearest Neighbors (KNN), Support Vector Classifier (SVC), Decision Tree, Random Forest, Bagging, AdaBoost, Naive Bayes, Gradient Boosting, Multi-Layer Perceptron (MLP), Stochastic Gradient Descent (SGD), LightGBM, CatBoost, and XGBoost were implemented and evaluated.

The performance of these models was compared based on accuracy metrics for different ratios of train-test splits to determine which one was the most effective in prediction. The results show that ensemble and boosting methods perform better than the rest due to their capacity to manage complex relationships in the data. The study also carried out correlation analysis to ascertain factors that affect depression. This demonstrates how academic and work-related stress contribute to depression in students.

The study demonstrates the viability of using machine learning in predicting depression among students and thus lays a foundation for building predictive models for depression. This would be instrumental in preventing depression among students through actions taken by educational institutions and health workers

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Abbreviations

KNN	:	K-Nearest Neighbors
SVC	:	Support Vector Classifier
Light GBM	:	Light Gradient Boosting Machine
MLP	:	Multi - Layer Perception
SGD	:	Stochastic Gradient Descent
XGBOOST	:	eXtreme Gradient Boosting
ML	:	Machine Learning
CGPA	:	Cumulative Grade Point Average
DASS-21	:	Depression, Anxiety, and Stress Scale - 21 Items
PHQ	:	Patient Health Questionnaire
CTT	:	Classical Test Theory
IRT	:	Response Theory
EFT	:	Emotion Focused Theory
HRQOL	:	Health Related Quality of Life
MSM	:	Mindfulness Stress Management
PCOS	:	Polycystic Ovary Syndrome
PS	:	Psychological Formulation
DSP	:	Deliberate Self-Poisoning

Chapter-1

1.1 Introduction

With the increasing number of students experiencing mental illness and depression, there is a growing concern among researchers globally regarding mental health among students. Students in today's world are exposed to social pressures, career uncertainty, academic pressures and life style changes, have a major influence on the mental health of students. Depression being one of the most prevalent mental health problems among students, and it has been discovered to have a negative influence on students' academic performance, emotional stability, social relationships, and overall quality of life. Hence, it is important to identify the most important indicators linked to depression among students.

Student mental illness is shaped by a set of demographic, academic, psychological, and lifestyle variables. Variables such as gender, age, city, and profession are significant demographic variables that can have an influence on mental health outcomes. Academic-related variables such as academic pressure, CGPA, satisfaction with studies, and work/study hours are important in determining the level of stress and emotional state of a student. Too much academic pressure and too little academic satisfaction can contribute to increased levels of psychological distress, which can lead to depression.

Apart from the academic considerations, lifestyle and behavioural parameters such as sleep, eating habits, and job satisfaction are also important for understanding the mental health trends among students. Lack of quality sleep, poor eating habits, and lack of satisfaction with academic or professional roles lead to emotional exhaustion and mental instability. Work pressure and financial problems are also significant external stress factors that can worsen anxiety and depression, particularly among students who are juggling academic and professional responsibilities.

Psychological risk factors such as suicidal ideation and mental illness in the family are robust predictors of depression. Students who have mental illness in their families may be more susceptible to depression due to environmental and genetic factors. In addition, the presence of suicidal ideation is a serious mental health issue that emphasizes the importance of early mental health care and interventions in the educational setting.

The things that can affect students mental health are important to look at. We want to know what things make students more likely to get depression. We are looking at how different things are related to depression. These things include who the students are, like if they are a boy or a girl how old they are and where they live. We are also looking at how school's affecting them like how much pressure they feel how well they are doing in school and if they like studying. We want to know about their life too like how much they sleep what they eat and how many hours they work or study. We are also looking at how they feel, like if they think about hurting themselves if they worry about money and if someone in their family has had health problems.

If we can understand what things make students more likely to get depression, we can make plans to help them. We can make tests to see if students are at risk and we can make support plans for schools. If we can find out who is, at risk early teachers, counsellors and people who

make laws can help them. They can give them help teach them about mental health and provide services to support them. We can use this information to help students with depression.

This study looks at things about students, such as where they come from what they study, how they live and how they think, to figure out how depressed they might be. The goal of this study is to find out what things can tell us if a student is likely to have health problems. This study wants to help teachers and schools find ways to check if students are doing okay and to stop mental health problems from happening in the first place. The study is really about students and depression levels among students. It wants to know more, about students and how to keep students healthy.

1.1.1 Depression as a Target Variable in Student Mental Health Studies

Depression is widely recognized as a major mental health variable in student psychological research. It serves as a measurable indicator of overall mental well-being and emotional stability. In data-driven mental health studies, depression is often treated as the target variable, while various demographic, academic, and lifestyle factors are considered feature variables that influence its occurrence.

Depression is analysed as the dependent variable influenced by multiple independent variables such as gender, age, academic pressure, work pressure, sleep duration, financial stress, and family history of mental illness. By examining the relationship, it becomes possible to identify high-risk groups and develop predictive models for early mental health intervention.

Early prediction of depression among students is highly beneficial as it enables educational institutions and mental health professionals to provide timely counselling, support systems, and preventive strategies. This approach also contributes in reducing of stigma associated with mental illness and also promotes a healthier academic environment.

Variables such as gender, age, profession play a significant role in influencing mental health outcomes among students. Many researches indicates that mental health differ across genders due to many factors such as biological, social, and emotional. In many cases, female students report higher levels of emotional stress and depressive symptoms, while male students face pressure from expectations and career responsibilities.

Age is another important factor because students of different ages deal with different amounts of academic and mental stress. Younger students may have trouble adjusting to school, while older students may feel anxious about their careers and the pressure of their jobs. The city or place of residence also affects mental health because of differences in lifestyle, educational infrastructure, social support systems, and exposure to competitive environments. Balancing school and work responsibilities can make you feel emotionally drained and more likely to get depressed.

1.1.2 Academic Factors as Indicators of Depression

Variables related to academic is one of the strongest predictors of mental illness and depression in students. Variables such as academic pressure, CGPA, study satisfaction, and work/study hours have a significant impact on the mental condition of a student. Academic pressure, if high, leads to many mental illnesses such as chronic stress, anxiety, and emotional burnout.

CGPA, an important variable, acts as an indicator of a student's performance in their studies, which directly impact a student's mental confidence and self-esteem. Students who do not perform well often feel inadequate, fear of failure, and compare themselves to their companions, leading to depression. On the other hand, too much focus on maintaining a high level of performance in studies can lead to mental pressure and anxiety in a student.

Study satisfaction, an important variable, indicates a student's level of engagement and emotional connection with their studies. Low study satisfaction indicates a student's lack of motivation, disinterest, and academic frustration, all of which lead to a high level of depression in a student. Excessive work/study hours leave a student with little time to relax, play, and socialize, thereby affecting their mental condition.

1.1.3 Lifestyle and Behavioural Indicators of Mental Illness

Factors like sleep duration and dietary habits are major aspects of an individual's mental health. Sleep helps an individual to have better cognitive and emotional functions. In addition, sleep helps an individual to have better psychological well-being. Poor sleep duration and irregular sleep patterns have been found to have a strong association with stress, anxiety, and depression among students.

Dietary habits have been found to influence mental health through physical and cognitive functions. Poor dietary habits, irregular meals, and nutrient deficiencies have been found to influence an individual's mental health. Poor dietary habits among students have been found to influence emotional instability and depression.

Moreover, excessive screen time and lack of physical exercise have been found to influence mental health. These behavioural aspects of an individual show the importance of managing an individual's lifestyle to prevent depression.

1.1.4 Psychological Risk Factors and Family Influence

Psychological factors such as suicidal thoughts and family history of mental illness are among the most significant predictors of depression. The presence of suicidal thoughts among individuals is a significant warning signal for depression. It is a psychological indicator of a severe level of emotional distress and is a significant predictor of depression.

Family history of mental illness is a significant predictor of depression among students. This is because genetic predisposition and environmental factors contribute greatly to the

development of mental health issues. For example, students with a family history of depression and other psychological problems may be at a higher risk of developing the same problems.

Financial stress is a significant psychological factor among students, especially those who seek higher education. Financial stress is a significant psychological factor among students because they may be required to pay fees and other expenses.

1.1.5 Importance of Identifying Depression Indicators in Students

The identification of depression indicators among students is vital in order to ensure early diagnosis, prevention, and intervention in mental health. Educational institutions have a vital role to play in ensuring that the environment is conducive to promoting mental health among students. By studying different depression indicators such as academic pressures, sleeping patterns, financial pressures, and lifestyle, it becomes easier to identify depression among students and ensure that they are given the necessary counselling.

The identification of depression among students at an early stage is vital in order to ensure that the following are achieved:

- Severity of mental illness is reduced
- Academic performance is improved
- Emotional strength is built
- Suicidal tendencies are prevented
- Overall well-being among students is promoted

The identification of mental health indicators also becomes useful in ensuring that predictive models are developed to help in the automation of mental health screening.

1.1.6 Research Gap and Need for the Study

Many studies have been conducted on depression among students; however, few studies have considered limited aspects from the psychological or academic points of view. There is a lack of research that considers multiple aspects in a unified analytical framework. Furthermore, with the introduction of advanced data analytics and machine learning techniques, there is a growing need to develop models that could accurately predict depression based on multiple features. The present research aims to bridge this research gap by considering a variety of feature variables such as Gender, Age, City, Profession, Academic Pressure, Work Pressure, CGPA, Study Satisfaction, Job Satisfaction, Sleep Duration, Dietary Habits, Degree, Suicidal Thoughts, Work/Study Hours, Financial Stress, Family History of Mental Illness, etc., along with Depression as the target variable.

1.1.7 Significance of the Study

The present research has great importance in the field of education, psychology, and data science. By identifying the major indicators of depression among students, this research could

play a vital role in the development of effective student mental health monitoring systems and counselling approaches. Such identification could help educators and researchers in the development of effective approaches to improve the lives of students

Moreover, the utilization of multiple indicators in the model could help in the accuracy of depression detection and could assist in the development of depression detection models. This research could also play a vital role in promoting mental health awareness and seeking help among students to overcome depression and other health issues. This helps to lessen the stigma that comes with mental health issues.

In conclusion, mental illness and depression among students have become a critical issue that needs to be addressed in the modern educational environment, given the rising levels of academic, mental, and lifestyle-related stress factors. Depression, as the target variable, is affected by numerous factors, including the students' demographic factors, academic results, work-related pressures, lifestyle, financial pressures, and mental history, among others. These factors can be referred to as the indicators.

In this respect, the main objective of the research was to carry out a comprehensive analysis of the mental illness and depression among students, using multiple feature variables to predict the level of depression among students. In essence, the research was conducted to help create a mental health assessment system that promotes a healthy learning environment.

1.2 Background of the Study

Mental health being one of the most prominent health concerns in recent years, students who have to face various challenges in their academic and social life. The need to perform well in academics, have high grades, and manage their finances and future career prospects is one of the most significant factors that have affected students' mental health in various ways. Thus, various mental health concerns such as stress, anxiety, and depression have become common in learning institutions.

However, among various mental health concerns, depression is one of the most severe and prevalent mental health concerns among students in various parts of the world. Depression is a mental health disorder that is linked with feelings of sadness, loss of motivation, fatigue, sleep problems, and problems in concentrating and taking interest in various activities in one's daily life. The impact of depression has a severe effect on students' academic performance and overall life. If not diagnosed and treated in its early stages, depression can lead to various negative consequences such as emotional instability, poor academic performance, and suicidal thoughts.

Students suffer from depression due to academic, personal, social, and environmental factors. Academic pressure being one of the main factors that contribute to mental stress among students. The competitive academic environment usually requires students to perform well academically, undergo constant assessment, and meet tight deadlines, which cause mental stress. Variables such as CGPA, study satisfaction, academic pressure, and work/study hours are important factors that contribute to students' mental health.

Apart from academic pressure, work-related stress is also common among students who work part-time or pursue professional training alongside their academic responsibilities. Work pressure and job satisfaction can impact emotional stability and mental health of the student. Students who suffer from high work pressure and academic workload experience fatigue, burnout, and mental distress.

Lifestyle factors are one of the important determinants . Variables such as sleep duration and eating habits are strongly associated with physical and mental health. Inadequate sleep and unhealthy eating habits can have negative impacts on cognitive functioning and emotional stability, which can contribute to depressive symptoms. Similarly, long work or study hours can result in less time for relaxation and increased levels of stress.

Demographic and social factors have an influence on the mental health outcomes among students. Factors such as gender, age, city, profession, and degree have an influence on how an individual experiences stress and manages it. Lastly, financial stress is another significant factor among students, especially for those who bear the responsibility for managing educational costs as well as other expenses.

Psychological risk indicators such as family history of mental illness and suicidal thoughts are significant factors in understanding depression among individuals. Family history of mental illness is a significant factor since an individual who comes from a family that has a history of mental illness is more likely to develop depression. Suicidal thoughts are an indicator of extreme emotional distress, emphasizing the need for early mental health detection and intervention.

With the emergence of data science and machine learning, researchers are using computers to study mental health and identify patterns related to depression. Machine learning algorithms have the capability to handle large datasets and identify relationships between multiple variables, which might not be possible using traditional statistical methods. These machine learning algorithms have the capability to develop predictive models that can be used to identify students who might have a higher chance of developing depression.

The data set, which is used to identify the depression indicators in students, is taken from Kaggle and has information about 27,870 students in 18 columns. There are various columns in the data set, namely Gender, Age, City, Profession, Academic Pressure, Work Pressure, CGPA, Study Satisfaction, Job Satisfaction, Sleep Duration, Dietary Habits, Degree, Suicidal Thoughts, Work/Study Hours, Financial Stress, Family History of Mental Illness, etc., which affect the target variable. All these features are used as feature variables, while Depression is used as a target variable for prediction.

To compare the relationship between variables and depression, machine learning classification algorithms have been used in this study. Different classification algorithms have been employed in this study. Different classification algorithms used in this study include: Logistic Regression, K-Nearest Neighbors (KNN), Support Vector Classifier (SVC), Decision Tree, Random Forest, Bagging, Ada Boost, Naive Bayes, Gradient Boosting, Multi-layer Perceptron, Stochastic Gradient, Light Gradient, Cat Boost, and XGBoost. These classification algorithms

have been used to create a predictive system that is capable of classifying whether a student is suffering from depression or not based on the given variables.

In this study, various machine learning classification algorithms have been used to compare the performance of each algorithm in order to predict student depression. The predictive model developed using machine learning algorithms helps to understand the impact of various factors on student depression.

Understanding depression indicators is very important for developing an effective mental health support system within educational institutions. The early prediction of depression among students helps educators, policymakers, and counsellors develop an effective support system for students, promoting their well-being.

Therefore, this research focuses on analyzing student mental illness and depression indicators using a large data set and machine learning techniques in order to improve the understanding of student mental health patterns and contribute to the development of predictive systems for the early detection of depression among students.

1.3 Objectives

The primary objective of this study is to examine the indicators for mental illness and depression among students and develop a predictive system that can identify students who are at a higher risk of developing depression. Mental illness among students has become a major concern due to increasing academic pressure, changes in lifestyle, financial pressures, and social demands. It is important to understand the various factors that lead to depression among students. The primary objective of this study is to examine various demographic, academic, lifestyle, and psychological factors that might influence depression among students.

One of the primary objectives of this study is to identify the major factors associated with depression among students. The study examines various variables such as gender, age, city, profession, academic pressure, work pressure, CGPA, study satisfaction, job satisfaction, sleep duration, dietary habits, degree, suicidal thoughts, work or study hours, financial stress, and family history of mental illness. The study aims to understand the various factors that influence depression among students by analyzing the various variables mentioned above. The study aims to understand how various aspects of a student's life influence depression.

Another purpose of this study is to investigate the relationship between academic factors and depression. Academic performance factors such as CGPA, academic pressure, and study satisfaction may have a significant impact on students' emotional well-being. High academic pressure and long study hours may result in emotional exhaustion, which may lead to depression in students. Thus, this study aims to investigate the relationship between academic conditions and students' mental well-being.

Another purpose of this study is to investigate the impact of lifestyle factors on students' mental well-being. Lifestyle factors such as sleeping hours and dietary habits may have a significant impact on physical and mental well-being in students. Sleeping hours and dietary habits may

result in emotional instability and mood swings, which may lead to depression in students. Thus, this study aims to investigate the impact of lifestyle factors on depression in students.

Another important aim of this research is to analyze various psychological and socio-economic factors that might influence student mental health. Financial problems are one of the common problems faced by many students. These problems might lead to more anxiety and pressure on their minds. In a similar manner, if any student has a family history of mental illness, they might be more prone to depression due to various environmental factors. Suicidal thoughts are one of the important factors that might influence depression in a significant manner.

Apart from analyzing individual factors, this research aims to understand how various factors influence depression in a combined manner. It is a fact that mental health problems are often influenced by a combination of academic, personal, lifestyle, and environmental factors. Hence, this research aims to provide a comprehensive analysis of various factors that influence depression among students.

Another major objective of this research is to utilize machine learning algorithms for the prediction of depression among students based on the given indicators. The dataset used for this study is obtained from Kaggle, which contains 27,870 student records with multiple attributes related to students' demographic, academic, lifestyle, and psychological information. These machine learning algorithms help to develop predictive models that classify whether a student is likely to have depression or not.

Machine learning algorithms used for this study include Logistic Regression, K-Nearest Neighbors (KNN), Support Vector Classifier (SVC), Decision Tree, Random Forest, Bagging, AdaBoost, Naive Bayes, Gradient Boosting, Multi-Layer Perceptron, Stochastic Gradient Descent, Light Gradient Boosting (LightGBM), CatBoost, and XGBoost, which are used to analyze the dataset and identify patterns that might be used for the prediction of depression among students.

Another objective of the study is to compare the effectiveness of these machine learning algorithms in predicting depression. It is also possible that different algorithms may vary in their accuracy and effectiveness in predicting the outcome when the same dataset is used. Therefore, by comparing different algorithms, it will be easy to identify the best algorithm that can be used in predicting depression based on the indicators used.

In conclusion, the aim of this research was to analyze the different factors that affect student mental health. Additionally, the research aimed to develop a machine learning-based model that can help in predicting students who may likely develop depression. The findings of this research may help in improving the mental health of students.

1.3.1 Purpose

The aim and objective of this project are to identify and analyze the various indicators that are associated with mental illness and depression among students and create a predictive model that is capable of accurately predicting students who might be at risk of developing depression. Depression is one of the most common types of mental illness among students in recent times

due to various factors such as academic pressure, lifestyle, financial problems, and social factors. All these factors have a significant impact on the lives of students. Hence, it is essential to understand various factors that influence depression among students and create methods that could assist in the prevention and early detection of depression among students.

This project aims to understand the relationship between various factors, including depression among students. The data used for this research is obtained from Kaggle. The variables include gender, age, city, profession, academic pressure, work pressure, CGPA, study satisfaction, job satisfaction, sleep duration, dietary habits, degree, suicidal thoughts, work or study hours, financial stress, family history of mental illness, etc.

Another significant purpose of this project is to apply machine learning algorithms to analyze the data and predict the level of depression among students using the provided indicators. Machine learning algorithms have the ability to process large amounts of data and uncover hidden patterns that may not be visible using other analysis techniques. By using different classification algorithms, this project aims to develop models that can be used to identify students who are at a higher risk of depression.

Another purpose of this project is to compare the efficiency of different machine learning algorithms in predicting depression among students using the provided indicators. By comparing different models, this project aims to determine which algorithm is most effective in predicting depression among students.

The primary aim of this undertaking is to furnish a data-informed approach for scrutinizing the determinants influencing student mental health, alongside the development of a predictive model designed to facilitate the early identification of student depression. The outcomes of this investigation hold the potential to aid educational establishments, counselling professionals, and policymakers in the formulation of impactful mental health support frameworks, counselling initiatives, and awareness endeavours. These can contribute to the enhancement of student welfare and the mitigation of depression within the academic environment.

1.3.2 Scope

The research scope of this study is to analyze the factors related to mental illness and depression among students. In recent years, mental health problems among students have been rising due to various academic, social, and personal pressures. The purpose of this study is to investigate the various signs that may lead to depression and how these signs affect the mental health of students.

The research study takes into consideration various demographic, academic, lifestyle, and psychological factors that may impact depression among students. These factors include gender, age, city, profession, academic pressure, work pressure, CGPA, study satisfaction, job satisfaction, sleep duration, dietary habits, degree, suicidal thoughts, work or study hours, financial problems, and family history of mental illness. By studying these factors, the research study will help to understand how various aspects of a student's life may lead to mental health problems and depression.

The data set used in this research is collected from Kaggle and consists of a large number of data points for students, which enables a detailed analysis of the relationships between various indicators and depression. The variable “Depression” is taken as the target variable, which indicates the mental health status of the students. The objective of this research is to analyze the data set and determine the patterns and relationships between the variables chosen and depression.

The research is limited to the analysis of the data set and the factors that are included in it. The research only considers the factors that are included in the data set and does not consider any other factors that may affect depression, such as psychological factors. The research is also limited to the analysis of the data set and does not involve the diagnosis or treatment of mental health disorders.

The results of this investigation provide significant perspectives for educational establishments, counselling professionals, and those involved in policy-making, thereby augmenting their comprehension of the factors influencing student mental health. By identifying key indicators related to depression, this study will greatly help in the development of awareness campaigns, support programs, and preventive strategies, and thus will promote student well-being.

Furthermore, the study is also aimed at understanding the various academic and lifestyle conditions that affect the mental health of students. The academic workload, satisfaction with the study, and educational pressures are likely to affect the mental stability of the students to a certain extent. The study is also aimed at bringing into focus the significance of maintaining an academic environment that is balanced for the educational as well as mental well-being of the students.

The study is also aimed at understanding the personal as well as environmental influences that affect the mental health conditions of the students. The financial pressures, family backgrounds, and lifestyle are likely to affect the mental stability of the students to a certain extent. This could cause stress, which may further lead to depression in the long run. The significance of providing support for the mental health of the students may also be understood in this context.

The relevance of this study is based on the fact that it seeks to investigate the various indicators associated with mental illness and depression among students using the available dataset.

Lastly, the study aims to contribute to the general understanding of various mental health issues affecting students through insights gained from the available data. The analysis of depression indicators among students may help in better understanding and identifying the risk factors associated with depression, and developing strategies that promote healthy lifestyles and better management of mental health among students.

Chapter-2

2.1 REVIEW OF LITERATURE

A cross-sectional study among 1,412 school-going adolescents in urban Patna, Bihar, using the Beck's Depression Inventory-II to assess the prevalence of depression. The study found a high overall prevalence of depression (49.2%), with 7.7% having severe depression. Depression was significantly more common among girls, older students, and those belonging to minority religious groups. The most commonly reported symptoms included guilty feelings, pessimism, sadness, and past failure. The authors concluded that adolescent mental health is largely neglected and emphasized the need for greater awareness among parents and teachers for early identification and support of depressed students in schools [1]. A study among 413 medical students at the University of Tartu to examine the relationship between sleep problems and symptoms of anxiety and depression using the Emotional State Questionnaire (EST-Q). The study found that 21.9% of students had symptoms of anxiety and 30.6% had symptoms of depression, with higher prevalence among females. Regression analyses showed gender-specific associations between sleep disturbances and emotional symptoms: in women, anxiety was linked to nightmares and morning fatigue, while depression was associated with insomnia, nightmares, nocturnal eating, and daytime sleepiness; in men, depression was mainly related to difficulty falling asleep before exams and poor subjective sleep quality. The authors concluded that a substantial proportion of medical students experience emotional distress and that specific sleep problems may serve as indicators of underlying anxiety and depression [2].

A survey among 190 senior high school students in Kolkata to examine academic stress, parental pressure, and mental health using a structured questionnaire and the General Health Questionnaire. The study found that nearly two-thirds (63.5%) of students experienced academic stress, and a similar proportion (66%) reported significant parental pressure for better performance. Parental pressure varied according to parents' education, mother's occupation, number of private tutors, and students' academic performance, with higher pressure reported among students whose fathers were non-graduates. About 32.6% of students showed signs of psychiatric distress, and 81.6% reported examination-related anxiety. The authors concluded that academic stress is a major mental health concern for high school students and emphasized the need for strategies to reduce academic pressure and improve psychological well-being [3]. Compared the screening performance of five commonly used questionnaires—GHQ-12, PHQ-9, K10, K6, and SRQ-20—for detecting common mental disorders in primary care settings in India. The study, conducted among 598 adult participants using the CIS-R as the diagnostic standard, found that all questionnaires demonstrated moderate to high discriminating ability and good internal consistency, with GHQ and SRQ showing the best performance. The tools were moderately to highly correlated with each other, although the positive predictive values ranged from 51% to 77% at optimal cut-off scores. The authors concluded that while all instruments are reasonably accurate for screening, none achieved high predictive value without compromising sensitivity, and the choice of cut-off scores should be adapted to specific clinical and resource settings [4].

Investigated anxiety, stress, depression, and psychosocial functioning among 1,812 Indian adolescents aged 12–19 years using the DASS-21 and Strengths and Difficulties Questionnaire (SDQ). The study validated the three-factor structure of DASS and two-factor model of SDQ in both English and Hindi versions. Results indicated that late adolescents showed higher difficulties scores, while early adolescents demonstrated higher prosocial behaviour, and females scored higher on prosocial behaviour than males. Differences were also observed based on residence, type of school, and school attendance status. Negative family relationships and

poor self-concept were significantly associated with higher depression, stress, and psychosocial difficulties. The authors concluded that sociodemographic and psychosocial factors play a crucial role in influencing mental health and functioning among Indian adolescents [5]. Assessed stress, anxiety, and depression among undergraduate medical students using the DASS-21 scale through a self-administered questionnaire. The study reported high levels of psychological distress, with depression (55%) and anxiety (64%) being more prevalent than stress (34%), and female students exhibiting higher levels of anxiety and depression. The findings indicated a substantial mental health burden among medical students, highlighting the need for preventive strategies and mental health support services within medical education [6].

A cross-sectional study among 400 students preparing for competitive examinations in central India using the DASS-21 to assess depression, anxiety, and stress levels. The study found notable levels of psychological distress, with 20.3% showing moderate and 8.8% severe depression, 17.8% moderate and 27.3% severe anxiety, and 14.8% moderate and 12% severe stress. Stress was significantly higher among females and students with more than two siblings, while anxiety and depression were strongly associated with non-supportive teachers and parents, inadequate sleep, and limited study time. The authors concluded that interpersonal support and lifestyle factors play a crucial role in students' mental health and emphasized the need for better communication and awareness among parents and teachers to reduce psychological distress [7]. A confirmatory factor analysis of the DASS-21 among a large community sample of Indian adults to examine its factor structure. The findings showed that both the three-factor model (depression, anxiety, and stress) and a single-factor model of general psychological distress demonstrated good model fit, with very strong correlations among the subscales ($r \geq .80$). This indicated substantial overlap between the constructs measured by the DASS-21. The authors concluded that the scale largely assesses general psychological distress rather than clearly distinct dimensions of depression, anxiety, and stress in the Indian adult population [8].

A school-based cross-sectional study among adolescents in Delhi using the DASS-21 to assess the prevalence of depression, anxiety, and stress. The study reported high prevalence rates of depression (47.9%), anxiety (65.3%), and stress (51.8%), with most students experiencing moderate depression, moderate anxiety, and mild stress. Psychological distress was more common among females, late adolescents, students living away from family, those from single or separated parents, alcohol users, and students facing family pressure to perform academically. The authors concluded that the high levels of mental health problems among adolescents highlight the need for counselling, parental support, and extracurricular activities to promote a healthier school environment [9]. a cross-sectional study among 2,798 professional course students in medical, dental, and engineering colleges in Patna using the Beck's Depression Inventory-II to assess depressive symptoms. The study found a high overall prevalence of depressive symptoms (47.78%), with the highest prevalence observed among engineering students, followed by dental and medical students. The findings indicated that depression significantly affects students' academic performance, interpersonal functioning, and professional development. The authors concluded that the alarming prevalence of depression among professional students necessitates early screening, diagnosis, and the establishment of psychological support services within educational institutions [10].

A systematic review and meta-analysis of 13 cross-sectional school-based studies published between 1990 and 2020 to estimate the pooled prevalence of depression among school-going adolescents in India using the Beck's Depression Inventory. The study reported a high pooled prevalence of depression at 53% (95% CI: 41%–65%), with higher prevalence among females

(57%) compared to males (50%). Subgroup analysis showed that prevalence increased with educational level, being highest among pre-university students. The authors concluded that more than half of Indian adolescents experience depression ranging from mild to severe and emphasized the need to strengthen school-based mental health services, public health policies, and community-level interventions for effective prevention and management [11]. Examined the relationship between psychological variables and depression among 500 school and college students aged 14–17 years. The study assessed depression, perceived stress, coping strategies, hardiness, and self-efficacy using standardized psychological scales. Findings indicated that depression was significantly related to levels of perceived stress, coping patterns, hardiness, and self-efficacy among adolescents. Factor analysis was used to identify the underlying relationships between these psychological variables and depressive symptoms. The authors highlighted that individual psychological resources and coping mechanisms play a crucial role in influencing depression among school and college students [12].

A cross-sectional study among 200 MBBS students in a medical college in Bengaluru using the DASS-21 to assess depression, anxiety, and stress. The study reported a high prevalence of psychological distress, with 48% of students experiencing depression, 54.5% anxiety, and 31% stress, with the majority of participants aged 18–22 years and a higher proportion of females. The findings indicated that medical education-related stress significantly affects students' mental health and may influence academic performance and professional functioning. The authors concluded that psychological morbidity among medical students is a neglected issue and emphasized the urgent need for accessible counselling services and mental health support within medical institutions [13]. A web-based cross-sectional study to examine the mental health status of college and university students in India during the first and second waves of the COVID-19 pandemic. The study found a significant increase in depression, anxiety, and stress during the second wave compared to the initial phase of the pandemic. Major contributing factors included difficulties in adapting to online learning, fear of educational disruption due to financial constraints, reduced physical interaction, and prolonged screen time. The authors concluded that the pandemic had a substantial negative impact on students' mental health and recommended counselling interventions and coping strategies, along with follow-up studies to assess long-term psychological effects of prolonged home confinement [14].

A psychometric validation study of the Physical Health Questionnaire (PHQ) among 101 Indian college students aged 18–23 years using exploratory and confirmatory factor analysis. The findings revealed a four-factor structure explaining 62.25% of the cumulative variance, along with satisfactory internal consistency and convergent validity. Confirmatory factor analysis further supported the model with good fit indices, indicating the robustness of the scale. The authors concluded that the PHQ is a reliable and psychometrically suitable instrument for assessing physical health and well-being among Indian college students and young adults [15]. A cross-sectional web-based study among 324 college students in India to examine the impact of the COVID-19 pandemic on mental health. The study reported that 68.8% of students experienced high fear of COVID-19, 28.7% had moderate to severe depression, and 51.5% had mild to severe anxiety. Having an infected family member was significantly associated with higher anxiety and depression, and the Fear of COVID-19 Scale showed moderate positive correlations with GAD-7 and PHQ-9 scores. The authors concluded that the pandemic created a substantial mental health crisis among college students and highlighted the need for psychological support and intervention strategies [16].

Investigated the prevalence and correlates of depressive symptoms among high school students in Hanover, Jamaica, using an abbreviated Beck's Depression Inventory-II. The study found

that 14.2% of adolescents reported depressive symptoms. Factors such as sexual activity, parental monitoring, and levels of maternal and paternal affection and support were significantly associated with depressive symptoms, with lack of maternal affection emerging as the strongest predictor. The authors emphasized the importance of family support systems and early screening to reduce the risk of depression among adolescents [17]. This cross-sectional study among 308 pre-university students in Mangalore reported a very high prevalence of depression (79.2%), with the majority experiencing moderate levels of depression. The study also found that both prevalence and severity of depression increased with age and were significantly higher among commerce students compared to science students, while no significant association was observed with gender or type of college. The authors emphasized the importance of mental health awareness, early identification, and the establishment of counselling services in educational institutions to address the high burden of depression among students [18].

An analysed three annual waves of longitudinal data from girls aged 9–15 years in the Great Smoky Mountains study to examine the link between puberty and depression. Their findings showed that when hormonal changes were included in the model, the apparent effect of Tanner stage disappeared, indicating that biological changes were more important than physical pubertal appearance. Specifically, variations in testosterone and oestradiol were significantly associated with depression, whereas FSH and LH showed no effect. The study concluded that the increased risk of depression in adolescent girls is more likely related to hormonal changes rather than morphological changes of puberty or their psychosocial impact [19]. A school-based study among 242 affluent adolescent students and found a considerable prevalence of depression, anxiety, and stress. The emotional burden was significantly higher among female students and those studying in board classes (10th and 12th), indicating greater academic pressure at these levels. Furthermore, all three conditions showed an inverse relationship with academic performance, and students who had experienced recent adverse life events were more likely to report higher levels of psychological distress, highlighting the role of both academic and personal stressors in adolescent mental health [20].

Evaluated the psychometric properties of the DASS among employees absent from work due to mental health problems and found high internal consistency and good construct and criterion validity. The scale demonstrated a clear three-factor structure and was effective in ruling out anxiety and depression after a negative test result, although it showed relatively low specificity following a positive result, suggesting its usefulness mainly as a screening rather than diagnostic tool in occupational settings [21]. examined the psychometric properties of the DASS using two large clinical samples and found the scale to be highly reliable and valid for clinical use. In the first study, the DASS showed excellent internal consistency and temporal stability, and exploratory factor analysis supported a three-factor structure consistent with previous nonclinical findings. The scale was also able to significantly differentiate between various anxiety and mood disorder groups. In the second study, confirmatory factor analysis further supported the latent three-factor structure of the DASS. Strong correlations with other established measures of anxiety, depression, and negative affect provided evidence for both convergent and discriminant validity, reinforcing the usefulness of DASS in assessing overlapping yet distinct features of anxiety and depression in clinical populations [22].

Analysed data from the large National Epidemiologic Survey on Alcohol and Related Conditions to compare the mental health of college students with their non-college-attending peers aged 19–25 years in the United States. The study revealed that nearly half of college-aged individuals had experienced a psychiatric disorder in the past year, with no significant

overall difference between college students and non-students. However, college students had a higher unadjusted risk of alcohol use disorders, although this difference disappeared after controlling for sociodemographic factors. They were less likely than their peers to have drug use disorders, nicotine dependence, or to use tobacco, and bipolar disorder was also less common among college attendees. Despite the high prevalence of psychiatric conditions, fewer than one-quarter of affected individuals sought treatment, highlighting a major gap in mental health care among young adults [23]. highlighted that college is a developmentally challenging period marked by academic pressure, separation from family, and multiple responsibilities, which often coincide with the first onset or worsening of mental health and substance use problems. The authors discussed the high prevalence of psychiatric disorders among college students and emphasized the importance of assessing age of onset of psychopathology. They also noted that many mental health problems tend to persist during college years, making early identification and intervention crucial. The paper further outlined key treatment considerations, including the involvement of parents, coordination with other healthcare providers, and the use of technology to improve treatment adherence, concluding that understanding the unique developmental and environmental context of college life is essential for effective mental health care [24].

A reviewed Indian study on depression among children and adolescents and reported wide variation in prevalence depending on study setting—ranging from 1.2%–21% in clinic-based studies, 3%–68% in school-based studies, and 0.1%–6.94% in community studies. The review highlighted that only one incidence study was available in India, estimating incidence at 1.6%. The authors identified multiple risk factors for depression, including academic difficulties, family and relationship problems, and economic stressors. They also noted that evidence on symptom profiles was limited, though common symptoms included low mood, loss of interest, aggression, sleep and appetite disturbances, and somatic complaints. The study concluded that methodological differences explain the wide prevalence range and that there is a significant gap in research on treatment effectiveness for childhood and adolescent depression in India [25]. examined the psychometric properties of the DASS-21 among 364 Ethiopian university students using both Classical Test Theory and Item Response Theory approaches. Factor analysis supported a one-factor structure representing general psychological distress rather than three distinct subscales. The scale demonstrated excellent reliability (Cronbach's α and McDonald's $\omega = 0.93$) and adequate divergent validity when compared with the Hamilton Anxiety Scale. Most items showed satisfactory psychometric performance, although two items displayed sub-optimal characteristics in both CTT and IRT analyses. Overall, the study concluded that the DASS-21 is a robust and reliable measure of psychological distress among Ethiopian university students, with minor item-level limitations [26].

A systematic review and meta-analysis of 17 studies involving 8,798 Ethiopian students to estimate the pooled prevalence of depression. The study found that the overall prevalence of depression was 35.52%, with an even higher prevalence in Addis Ababa (40.43%). Significant associated factors included female sex, being a first-year student, family history of mental illness, alcohol consumption, and parental neglect. The authors concluded that depression among Ethiopian students is considerably high compared to the general population, emphasizing the urgent need for strengthened mental health services and psychosocial support within educational institutions [27]. Evaluated the validity of the PHQ-9 as a brief measure of depression severity among 6,000 patients in primary care and obstetrics–gynaecology settings. The study demonstrated strong construct validity, as higher PHQ-9 scores were associated with poorer functional status, greater symptom-related difficulty, more sick days, and increased health-care utilization. Using a structured mental health professional interview as the gold

standard, a PHQ-9 cut-off score of ≥ 10 showed high sensitivity (88%) and specificity (88%) for major depression. The scale was also able to categorize depression severity into mild, moderate, moderately severe, and severe levels. The authors concluded that the PHQ-9 is a reliable, valid, and practical tool for both clinical diagnosis and research on depression severity [28].

A prospective cohort study among 154 first-year female medical students in South India to examine changes in depression, anxiety, and stress over the first six months of medical training. Although mean DASS-21 scores were within the normal range at baseline, there was a significant increase in psychological distress after six months of medical education, indicating a possible impact of the medical school environment on mental health. The study found that higher distress was positively associated with neuroticism and emotion-focused coping styles, suggesting that personality and coping strategies play an important role in students' mental well-being. Despite elevated distress levels, very few students sought professional help, highlighting a treatment gap. Academic performance was not significantly associated with psychological distress. The authors concluded that medical colleges need cultural changes and more accessible, student-friendly mental health services to address rising mental health concerns [29]. A systematic review and meta-analysis to examine the relationship between stress, time management, and academic performance in preclinical medical education. After reviewing 94 eligible studies, the authors found no evidence of publication bias and reported a significant negative pooled correlation ($r = -0.32$) between stress and academic performance, indicating that higher stress levels were associated with poorer academic outcomes. The review also highlighted time management as a key modifiable factor influencing academic success. The authors concluded that improving students' self-regulatory skills, such as stress management and time management, through targeted interventions could potentially enhance academic achievement and should be explored further through experimental research [30].

Systematically reviewed case-finding instruments used to detect major depression in primary care settings by analysing 18 studies involving more than 15,000 patients. The review compared nine different screening tools and found that, despite variation in length and administration time, all instruments showed similar diagnostic performance, with an overall sensitivity of 84% and specificity of 72%. The authors demonstrated that routine screening could identify most cases of depression in primary care, although some false positives would occur. They concluded that several reliable screening tools are available, and selection should be based on practical considerations such as ease of use, time required, and suitability for monitoring treatment response rather than on major differences in accuracy [31]. A 76-week randomized, double-blind controlled trial to examine the effectiveness of maintenance treatment with sertraline in patients with chronic or double depression. The study found that patients who continued sertraline had a significantly lower rate of depression recurrence compared with those receiving placebo (6% vs. 23%). Clinically significant depressive symptoms reemerged in only 26% of patients on sertraline compared to 50% in the placebo group. Using Cox regression analysis, patients on placebo were over four times more likely to relapse even after adjusting for clinical factors. The authors concluded that long-term sertraline therapy is safe and highly effective in preventing relapse in chronically depressed patients [32].

A large web-based survey among 2,843 undergraduate and graduate students at a public university to estimate the prevalence of depression, anxiety, and suicidality. The study found that 15.6% of undergraduates and 13.0% of graduate students met criteria for a depressive or anxiety disorder, and 2% reported suicidal ideation in the previous four weeks. Students facing financial difficulties were significantly more likely to experience mental health problems, with

odds ratios ranging from 1.6 to 9.0. The authors concluded that university campuses represent a critical setting for mental health interventions, particularly for students from lower socioeconomic backgrounds [33]. A prospective study among 400 college students aged 18–25 years to assess the prevalence of depression and anxiety using Beck's Depression and Anxiety Inventory. The study found that 48.3% of students had depression and 50% had anxiety, indicating a high burden of mental health problems among college students. The authors concluded that a significant proportion of students experience moderate-to-severe levels of depression and anxiety, emphasizing the need for early clinical diagnosis and timely intervention [34].

A cross-sectional study among 822 university students in Turkey to assess the prevalence and correlates of depression and its impact on health-related quality of life (HRQoL). The study found that 21.8% of students had depression, with significant risk factors including family history of depression, acne, physical defects, smoking, alcohol use, and future occupational concerns. Students with depression scored significantly lower across all domains of the SF-36 quality-of-life scale, indicating poorer physical and mental well-being. The authors concluded that depression is widespread among university students and emphasized the need for mental health education and preventive programs to improve student well-being [35]. A cross-sectional comparative study among 150 medical and 150 engineering students to assess levels of stress, depression, and anxiety using standardized scales. The study found that a higher proportion of medical students experienced moderate to high stress (72%) compared to engineering students (56.7%). The prevalence of depression (20.6%) and anxiety (19.4%) was also greater among medical students than among engineering students (15.3% and 11.3%, respectively). The authors concluded that medical students are at greater psychological risk due to academic and professional pressures and recommended the early introduction of stress management and mental health support programs within medical education [36].

A cross-sectional comparative study among 150 medical and 150 engineering students to assess levels of stress, depression, and anxiety using standardized scales. The study found that a higher proportion of medical students experienced moderate to high stress (72%) compared to engineering students (56.7%). The prevalence of depression (20.6%) and anxiety (19.4%) was also greater among medical students than among engineering students (15.3% and 11.3%, respectively). The authors concluded that medical students are at greater psychological risk due to academic and professional pressures and recommended the early introduction of stress management and mental health support programs within medical education [37]. A cross-sectional study among medical students in Jhansi, Uttar Pradesh to assess the prevalence of depression and anxiety using the DASS scale. The study found that 64% of students experienced depression and anxiety, with most cases being of mild to moderate severity. Female students and first-year students showed higher prevalence of depression compared to their counterparts. Significant associations were observed between psychological morbidity and factors such as family problems, substance use, hostel stay, and family history of mental illness. The authors concluded that depression and anxiety are highly prevalent among medical students and emphasized the need for early screening and psychiatric counselling to address this growing mental health concern [38].

A large cross-sectional study among 1,617 Turkish university students using the DASS-42 to assess depression, anxiety, and stress. The study found that moderate or higher levels of depression were present in 27.1%, anxiety in 47.1%, and stress in 27% of students, with female students showing significantly higher anxiety and stress levels. First- and second-year students reported greater psychological distress compared to senior students, while those satisfied with

their education had lower levels of depression, anxiety, and stress. The authors concluded that the high prevalence of psychological distress among university students is concerning and highlighted the need for preventive mental health services and student support systems [39]. A cross-sectional study among 1,617 Turkish university students using the DASS-42 to assess levels of depression, anxiety, and stress. The study found that moderate or higher levels of depression were present in 27.1%, anxiety in 47.1%, and stress in 27% of students, with female students reporting significantly higher anxiety and stress. First- and second-year students experienced greater psychological distress compared to senior students, while students satisfied with their education showed lower levels of depression, anxiety, and stress. The authors concluded that the high prevalence of psychological distress among university students is alarming and emphasized the need for preventive mental health services and structured student support systems [40].

A systematic review of 40 studies published between 1980 and 2005 to examine depression, anxiety, and other indicators of psychological distress among U.S. and Canadian medical students. The review found consistently high levels of depression and anxiety, with psychological distress being greater than that of the general population and age-matched peers, particularly in later years of training. The authors also noted that distress tended to be higher among female medical students, although evidence on specific causes and academic consequences was limited. They concluded that medical training is a period of substantial psychological burden and emphasized the need for large, prospective, multicentre studies to better understand risk factors, outcomes, and implications for professional competence [41]. examined gender differences in depression among undergraduate students using three conceptualizations of depression—depressed mood, depressive syndrome, and depressive disorder—across two samples. The study found no significant gender differences in rates of depressed mood or depressive syndrome. However, when depression was assessed as a clinical disorder, male students were more likely than female students to be classified as depressed in both non-referred and counselling-seeking samples [42].

A systematic review and Bayesian network meta-analysis of 25 randomized controlled trials involving 1,453 women with PCOS to compare the effectiveness of different interventions for anxiety and depression. The study found that emotion-focused therapy (EFT) and peer support (PS) were the most effective interventions in reducing both anxiety and depressive symptoms, with strong rankings based on SUCRA analysis. Mindfulness stress management (MSM) and omega-3 plus vitamin E supplementation (O3+VE) also showed beneficial effects on emotional well-being, whereas metformin and vitamin D plus probiotics did not demonstrate significant mental health benefits compared with control conditions. The authors concluded that psychological and supportive interventions, particularly EFT and PS, should be prioritized in managing mental health symptoms in women with PCOS [43]. A large web-based survey among 7,915 first-year tertiary education students in Hong Kong using the DASS-42 to assess depression, anxiety, and stress. The study found that moderate or higher levels of depression were present in 21% of students, anxiety in 41%, and stress in 27%, indicating a substantial burden of psychological distress in the transition to college life. The authors noted that online data collection was well accepted by students. They concluded that the high prevalence of mental health symptoms in first-year students is alarming and highlighted the need for primary and secondary prevention strategies along with structured campus mental health support services [44].

Compared stress and depressed mood among medical, law, and graduate students at McGill University using the Derogates Stress Profile (DSP). The study found that although medical

students reported subjective feelings of stress slightly above population norms, their overall stress levels were actually lower than those of law students, graduate students, and the general population. A small proportion of medical students showed elevated depressive symptoms, but these did not appear to be directly linked to medical school stress. However, the transition from basic science training to clinical training was associated with a noticeable increase in stress and depressed mood. The authors concluded that medical students are not more stressed than other student groups overall, but that targeted interventions during critical transition periods in medical training may be beneficial for mental health [45]. A cross-sectional study among medical students in Years 1, 3, and 6 at Karolinska Institute to examine stress exposure and depressive symptoms using the Higher Education Stress Inventory and the Major Depression Inventory. The study found that different stressors were prominent at different stages of training: Year 1 students reported high workload and lack of feedback, Year 3 students were more concerned about future competence and pedagogical shortcomings, and Year 6 students additionally perceived a non-supportive climate. Female students consistently reported higher stress levels than males across most domains. The prevalence of depressive symptoms among medical students was 12.9%, significantly higher than in the general population, with rates of 16.1% in females compared to 8.1% in males. The authors concluded that medical training is associated with substantial stress—particularly in the first year—and that female students are at greater risk of both stress and depression [46].

A cross-sectional study among 93 first-year veterinary students at Kansas State University to assess depression and anxiety using the CES-D and MHI-A scales. The study found that 32% of students experienced clinically significant depressive symptoms, and many also reported elevated anxiety levels. Key predictors of psychological distress included homesickness, poor physical health, and unclear instructor expectations. The authors suggested that targeted interventions addressing these factors could help improve mental well-being among veterinary students [47]. Analyzed data from the Canadian Community Health Survey to examine factors associated with continuity and discontinuity of depressive disorders among young Canadians aged 15–24 compared with older age groups. The study found that 10.2% of youth had experienced depression in their lifetime, and that social support was the key factor distinguishing younger individuals from older groups. Young people reported higher stress levels than adults, and the combination of high stress, weak social networks, and poor social support strongly differentiated long-lasting depression from non-cases. Low community cohesion was also linked to new cases of depression. The authors concluded that prevention efforts should focus on reducing stress and strengthening social support and community connectedness to prevent onset and recurrence of depression among youth [48].

UK normative data for the DASS using a large non-clinical sample of 1,771 adults and examined its latent structure and validity. Confirmatory factor analysis supported a three-factor model corresponding to depression, anxiety, and stress with correlated errors among related items, showing good model fit (CFI = .93). The study found that demographic variables had only minimal influence on DASS scores, while the scale demonstrated excellent reliability and adequate convergent and discriminant validity when compared with other measures such as HADS, sAD, and PANAS. The authors concluded that the DASS is a reliable and valid instrument for assessing negative emotional states and that the provision of large normative data enhances its clinical utility in the UK context [49]. examined how gender roles, stress, and coping styles were related to depressive symptoms among first-year college students. The study found that masculinity and femininity were significant predictors of problem-focused coping, while femininity also predicted emotion-focused coping. Higher levels of family and college-related stress, along with greater use of avoidant coping strategies, were strongly associated

with increased depressive symptoms. The authors concluded that understanding the interplay between gender roles, stress, and coping is important for supporting freshmen during their transition to university life and improving their psychological adjustment [50].

2.2 ABOUT THE DATA

2.2.1 Introduction

Data is significant in analyzing and understanding patterns in relation to mental health and depression issues faced by students. In this study, a structured dataset is used to investigate various demographic, academic, lifestyle, and psychological factors that may affect depression in students. The dataset gives a comprehensive overview of various attributes that are student-related and allows a detailed analysis of various factors that may lead to mental illness in students. This chapter gives a detailed description of the dataset in relation to the study.

2.2.2 Data Source

This study uses a dataset from Kaggle, a well-known site for datasets. Researchers and groups from many different fields share their high-quality datasets on Kaggle, which is a major hub. The dataset we chose is about student mental health and includes information about different things that could make students depressed.

There are 27,870 entries in this dataset, which are spread out over 18 columns. Each entry is for one student and has information that is different from the others. Because the dataset is so well-organized, it's easier to look for patterns and see how different things are related to depression.

2.2.3 Dataset Overview

There are both independent variables (features) and a dependent variable (target) in the dataset. The independent variables show different parts of a student's life, and the dependent variable shows whether the student is depressed.

- There are 27,870 records in total
- There are 18 columns in total.
- The target variable is depression.

The dataset has a good mix of qualitative and quantitative data, which makes it possible to look at mental health indicators in a full way. The variety of variables aids in comprehending the intricate nature of depression among students.

2.2.4 Description of Variables

The data set contains the following variables:

1. Gender

The variable represents the gender of the student. A categorical variable that may affect the mental health of students based on social and biological factors.

2. Age

This denotes the age group of the student. It is a numerical variable, which helps in understanding the variations in mental health among students belonging to different age groups.

3. City

This variable denotes the location or city in which the student lives. It helps in understanding the influence of environmental factors on the mental health of students.

4. Profession

This denotes the profession of the student, i.e., the current status of the student in terms of studies or work. It helps in understanding the level of stress experienced in a particular profession.

5. Academic Pressure

This variable denotes the level of academic pressure experienced by the student. It is generally believed that students facing high academic pressure are prone to depression.

6. Work Pressure

Work pressure refers to the amount of stress associated with the student's part-time work. It may cause the student to feel mentally fatigued and stressed.

7. CGPA

The Cumulative Grade Point Average (CGPA) represents the academic performance of the student. It is a numerical variable and may affect the self-esteem and mental health of the student.

8. Study Satisfaction

The study satisfaction variable represents the level of satisfaction associated with the academic life of the student. Low levels of satisfaction may cause the student to feel dissatisfied with academics and stressed.

9. Job Satisfaction

The job satisfaction variable represents the level of satisfaction associated with the academic or part-time work of the student. It may affect the emotional health of the student.

10. Sleep Duration

The sleep duration variable represents the number of hours the student sleeps. Sleep is important for the health of the individual.

11. Dietary Habits

This variable refers to the dietary habits of the student. Healthy dietary habits are linked with positive mental health outcomes.

12. Degree

This variable refers to the educational qualification or course pursued by the student.

13. Suicidal Thoughts

This variable refers to the suicidal thoughts experienced by the student. It is an important psychological variable for students.

14. Work/Study Hours

This variable refers to the number of hours devoted for work and study. Too much time devoted may cause stress and burnout.

15. Financial Stress

Financial stress refers to the financial pressure faced by the students. Financial stress is an important factor for students.

16. Family History of Mental Illness

This feature represents whether the student has a family history of mental illness. This feature may be a contributing factor to depression due to genetic and environmental factors.

17. Depression (Target Variable)

This is the target variable for this study. This feature indicates whether the student is suffering from depression or not. The aim of this study is to analyze this feature based on the other features.

2.2.5 Nature of the Data

The data set includes different data types, such as categorical and numerical variables, which offer a comprehensive representation of the student data.

List of Categorical Variables: Gender, City, Profession, Dietary Habits, Degree, Suicidal Thoughts, Family History of Mental Illness, Depression

List of Numerical Variables: Age, Academic Pressure, Work Pressure, CGPA, Study Satisfaction, Job Satisfaction, Sleep Duration, Work/Study Hours, Financial Stress

The data set includes different data types, making it easy to analyze the interrelationship between different variables and their effect on depression .

2.2.6 Data Quality and Reliability

The dataset obtained from the Kaggle platform is structured and organized. This dataset is appropriate for analysis. Nevertheless, just like other datasets in the real world, this dataset may have some challenges such as missing information or differences in the way information is represented. Therefore, the appropriate data preprocessing techniques are required for the analysis of the dataset.

The dataset is appropriate for use since it is reliable for academic and research purposes. This is because the dataset represents real-life scenarios with regard to students and depression. The dataset has diverse variables.

2.2.7 Limitations of the Dataset

Even though this data gives us significant insights into various aspects, it also has some limitations:

- The data is based only on the information provided in the dataset and may not include all the factors that affect depression
- No clinical diagnosis or medical evaluation is included in this data
- The data may be based on what a person answers in a survey and may be biased in some cases
- External factors such as culture and experiences are not fully included in this data

- These limitations must be kept in mind while analyzing the results obtained from this study.

2.2.8 Conclusion

In conclusion, the data set used in this research provides a comprehensive foundation for the analysis of student mental illness and depression indicators. The large number of records in the data set, i.e., 27,870 and the presence of relevant variables in the data set help in a comprehensive analysis of the factors that affect student mental health. In this chapter, an overall description of the data set used in this research is given. The source of the data set, the structure of the data set, the variables included in the data set, and the importance of the data set have been explained in detail.

Chapter-3

3.1 METHODOLOGY

3.1.1 Introduction

This chapter discusses the methodology used in analyzing student mental illness, as well as how depression is predicted using various indicators. In this study, a structured approach is used, which includes data collection, data preprocessing, exploratory data analysis, feature selection, modeling, and performance evaluation. The methodology is aimed at ensuring that the data is properly pre-processed, as well as analyzed, in order to generate useful insights regarding the factors that influence depression among students.

The data set used in the analysis is obtained from Kaggle, with 27,870 records on students, as well as 18 attributes. The attributes include demographic, academic, lifestyle, as well as psychological factors, while the “Depression” variable is used as the target variable, with the rest of the data used as the feature variable.

3.1.2 Research Design

The research methodology adopted for the research is quantitative and analytical. The research is based on data analysis. The data analysis is statistical and computational. The statistical and computational analysis is conducted to evaluate the correlation between different variables and depression.

The research design involves the application of various classification techniques. These classification techniques include the application of traditional machine learning algorithms as well as the application of advanced algorithms. In this study, the researcher has also included the application of Artificial Neural Networks (ANN) for the prediction of complex relationships.

The research design adopted for the research is as follows:

- 1.Data Collection
- 2.Data Preprocessing
- 3.Exploratory Data Analysis
- 4.Feature Engineering and Selection
- 5.Model Development
- 6.Model Evaluation
- 7.Results Analysis

3.1.3 Data Collection

The dataset used is collected from Kaggle, a popular website that is widely used. The dataset has 27,870 entries with various attributes concerning student demographic information, academic performance, lifestyle, and psychological conditions.

Each row in this dataset represents one student. The dataset has both categorical and numerical features that represent different aspects of student information.

3.1.4 Data Preprocessing

Preprocessing of data plays an important role in ensuring the quality of the data. The data collected may contain some missing values, inconsistencies, and may also contain some unwanted data.

The following steps are involved in the preprocessing of the data:

3.1.4.1 Dealing with Missing Values

If there are any missing values, the right methods are used to deal with them:

- Taking out data that has too many missing values
- Using the mean, median, or mode to fill in missing data

3.1.4.2 Cleaning the Data

Some of the steps in data cleaning are getting rid of duplicate data, fixing data inconsistencies, and making sure that the data is in the right format.

3.1.4.3 Putting Categorical Variables into Code

To turn data variables like gender, city, and dietary habits into numbers, you can use methods like:

- Encoding with labels
- One-Hot Encoding

3.1.4.4 Scaling Features

The numerical variables such as age, CGPA, and work/study hours are scaled so that all features contribute equally to the analysis.

3.1.4.5 Data Transformation

Data transformation is carried out to improve the variables' distribution, thereby improving the model's performance.

3.1.5 Exploratory Data Analysis (EDA)

Exploratory Data Analysis is carried out to understand the structure and nature of the given data set.

The following steps are carried out during Exploratory Data Analysis:

- Analysis of the data distribution
- Identification of outliers
- Correlation analysis
- Visualization using graphs and charts

Exploratory Data Analysis is helpful in understanding how various factors such as academic pressure, sleep hours, and financial stress are correlated with depression.

3.1.6 Feature Selection

Feature selection is done to select the most important variables that affect the student's depression. This helps improve the model.

The following aspects are taken into account during feature selection:

- Relevance of the variables with respect to the target variable
- Correlation between the variables
- Elimination of the variables that are not as important as the others

Important features that are taken into account are academic pressure, sleep duration, financial pressure, and suicidal thoughts, which are important with respect to the target variable.

3.1.7 Model Development

Model development is the process of developing a predictive model using the data that has been pre-processed. The data is divided into two types:

- Training Set: The data is used for developing the model
- Testing Set: The data is used for evaluating the model

Various classification algorithms are used to develop the model that helps predict whether the student is suffering from depression or not.

The models used in this study include:

- Logistic Regression
- K-Nearest Neighbors (KNN)
- Support Vector Classifier (SVC)
- Decision Tree
- Random Forest
- Bagging
- AdaBoost
- Naive Bayes
- Gradient Boosting
- Multi-Layer Perceptron
- Stochastic Gradient Descent
- Light Gradient Boosting (LightGBM)
- CatBoost

- XGBoost

Each model is trained using the training dataset and tested using the testing dataset.

3.1.8 Model Evaluation

The evaluation of the model is done in order to assess the performance of each model. The evaluation metrics used are:

3.1.8.1 Accuracy

Accuracy is the ratio of correct predictions made by the model.

3.1.8.2 Precision

Precision is the ratio of positive predictions that are correct.

3.1.8.3 Recall

Recall is the ratio of correct positive predictions.

3.1.8.4 F1-Score

F1-score is the harmonic mean of precision and recall.

3.1.8.5 Confusion Matrix

The confusion matrix is a table that shows the correct and incorrect predictions made by the model.

These metrics are used in comparing the different models and selecting the most efficient model.

3.1.9 Model Comparison

The different models are compared based on their performance. This is done in order to identify:

- The most accurate model
- The most reliable model
- The model with the most precise and accurate recall

This is done in order to ensure that the model selected is efficient in predicting the outcome.

3.1.10 Implementation Tools and Technologies

The tools and technologies used in this study are as follows:

- Programming Language: Python
- Libraries: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn
- Platform: Jupyter Notebook

These tools are used to carry out the analysis.

3.1.11 Limitations of Methodology

The methodology has some limitations as follows:

- The analysis is limited to the available data.
- The study does not take into consideration external factors that are not part of the available data.

3.1.12 Conclusion

The chapter described the methodology used to analyze student mental illness and predict depression. The methodology is based on a systematic approach that involves various steps. The study aims to identify various factors affecting depression using various techniques. The study also aims to develop an effective predictive system.

3.2 Methodology Used

3.2.1 Logistic Regression

Logistic Regression is a supervised learning algorithm that is used for binary classification problems. In this study, Logistic Regression is used to classify whether a student is suffering from depression or not based on various input features.

Unlike linear regression, Logistic Regression predicts probabilities instead of actual values using a sigmoid function. The output is always within the range 0 to 1.

The mathematical equation that is used by Logistic Regression is as follows:

$$P(Y = 1) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n)}}$$

Where:

Y is the dependent variable (Depression)

$X_1, X_2, \dots, X_n$ are independent variables

$\beta_0, \beta_1, \dots, \beta_n$ are coefficients

e is Euler's constant

Logistic Regression is widely used because it is simple, intuitive, and computationally efficient. Nevertheless, it assumes that features are linearly related to log-odds.

3.2.2 K-Nearest Neighbors (KNN)

K-Nearest Neighbors is a non-parametric supervised learning algorithm that is used for classification problems. KNN is a supervised algorithm that classifies new points based on the majority vote of their neighbors.

The steps that are followed by KNN are as follows:

1. Define K
2. Compute distance with all points
3. Select K points
4. Select majority

The distance is calculated as follows:

$$d = \sqrt{\sum (x_i - y_i)^2}$$

KNN does not make assumptions about data distribution. However, it is computationally expensive and sensitive to the choice of K.

3.2.3 Support Vector Classifier (SVC)

Support Vector Classifier is a supervised learning algorithm that finds the best hyperplane between two classes.

The goal of the algorithm is to find the maximum margin between the two classes. In the case of non-linear data, kernel functions are applied:

- Linear
- Polynomial
- Radial Basis Function

SVC works well in high-dimensional space, producing good results.

3.2.4 Decision Tree

Decision Tree is a supervised machine learning algorithm that uses a decision tree for decision-making. It works on the basis of splitting data according to the value of the features..

Decision Tree is a classification algorithm that works on a tree structure.

Structure:

- Root Node
- Decision Nodes
- Leaf Nodes

Decision Nodes are divided by:

- Gini Index
- Entropy

Decision Trees are simple, but they tend to overfit the data.

3.2.5 Random Forest

Random Forest is an ensemble method that makes predictions by combining the results of multiple decision trees. It is an improvement of the decision tree algorithm, which makes the prediction more accurate.

Steps:

1. Create subsets of data
2. Train decision trees
3. Aggregate the trees

Advantages:

- Reduces overfitting
- Improves accuracy
- Handles large data sets

3.2.6 Bagging

Bagging is an ensemble method that helps improve the performance of the model by creating multiple models, each of which is trained on a different data set. The predictions of the individual models are combined by averaging the predictions

Bagging, or Bootstrap Aggregating, helps in improving the stability of the model by:

- Sample the data with replacement
- Train multiple models
- Aggregate the predictions

It helps in reducing the variance of the model.

3.2.7 AdaBoost

AdaBoost is a boosting algorithm that helps improve the performance of the model by focusing on the data points that are misclassified by the model. The misclassified data points are given more importance, which helps improve the accuracy of the model.

AdaBoost helps in improving the performance of the model.

Steps:

1. Assign equal weights
2. Train the model
3. Assign weights to the misclassified samples
4. Combine the models

It helps in improving the performance of the model but is not suitable for noisy data sets.

3.2.8 Naive Bayes

Naive Bayes is a classification algorithm that works on the basis of Bayes' theorem. It is a probabilistic algorithm that assumes all features are independent of each other.

Naive Bayes is based on Bayes' Theorem:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

It assumes that all features are independent.

Advantages:

- Fast
- Efficient
- Suitable for large datasets

3.2.9 Gradient Boosting

Gradient Boosting is an ensemble algorithm that works on the concept of sequential models, which corrects the errors of the previous models. It works on the concept of minimizing the loss function for the purpose of achieving maximum accuracy.

Steps:

1. Training a base model
2. Calculating errors
3. Training another model on errors

Objective function:

$$Loss = \sum (y - \hat{y})^2$$

It is a powerful learner but computationally costly.

3.2.10 Multi-Layer Perceptron (ANN)

Multi-Layer Perceptron is an artificial neural network that works on the concept of multiple neurons in the network. It works on the concept of non-linear mapping of the data. It is used for deep learning, and it has the capability of achieving maximum accuracy.

Artificial Neural Network comprises:

- Input Layer
- Hidden Layer
- Output Layer

Neuron calculation:

$$Y = \sum (W_i X_i) + b$$

Activation functions:

- ReLU
- Sigmoid

ANN captures complex non-linear relationships.

3.2.11 Stochastic Gradient Descent (SGD)

Stochastic Gradient Descent is a type of optimization algorithm that is used to train models efficiently. The algorithm is used to update the parameters of a model using one data point at a time rather than using all the points.

Update rule:

$$\theta = \theta - \eta \nabla J(\theta)$$

Where:

η = Learning rate

SGD is computationally efficient when dealing with large datasets.

3.2.12 LightGBM

LightGBM (**Light Gradient Boosting Machine**) is another gradient boosting algorithm used to create models efficiently. The algorithm is known for being fast and is used to handle large datasets with high efficiency.

Features:

- Faster execution
- Less memory
- Leaf wise

Suitable for large datasets.

3.2.13 CatBoost

CatBoost is a gradient boosting algorithm that is used to handle categorical variables. The algorithm is known to reduce the need for preprocessing as well as improve the accuracy of models while reducing overfitting.

Advantages:

- No extensive preprocessing required
- Reduces overfitting
- High accuracy

3.2.14 XGBoost

XGBoost (**eXtreme Gradient Boosting**) is a sophisticated boosting algorithm. XGBoost is a boosting algorithm that is known to be advanced. The algorithm is known to be fast and is used in various competitions as well as in practical application. The algorithm has a feature that prevents overfitting.

Objective:

$$Obj = Loss + Regularization$$

Features:

- Regularization
- Parallel processing
- High performance

Chapter-4

4.1 Exploratory Data Analysis

Exploratory Data Analysis is the process of analyzing the data, exploring the data, etc., before applying machine learning algorithms on the data. EDA is the first and most important step in the data analysis process.

In this study, EDA is performed on the student mental health data set, analyzing the effect of various parameters, i.e., academic pressure, sleep duration, financial pressure, etc., on the mental health of students, i.e., depression.

Objectives of EDA:

- Data Cleaning
- Descriptive Statistics
- Data Visualization
- Feature Engineering
- Correlation & Relationship
- Data Segmentation
- Hypothesis Generation
- Data Quality Assessment

4.1.1 Data Cleaning

Data cleaning is the process of dealing with missing values, duplicates, etc., in the data set.

In this project, the following operations were performed:

- Missing values were detected and treated accordingly
- Duplicates were removed
- Data types were verified
- Outliers were detected

This step is important as the data has to be correct before moving on to the next step.

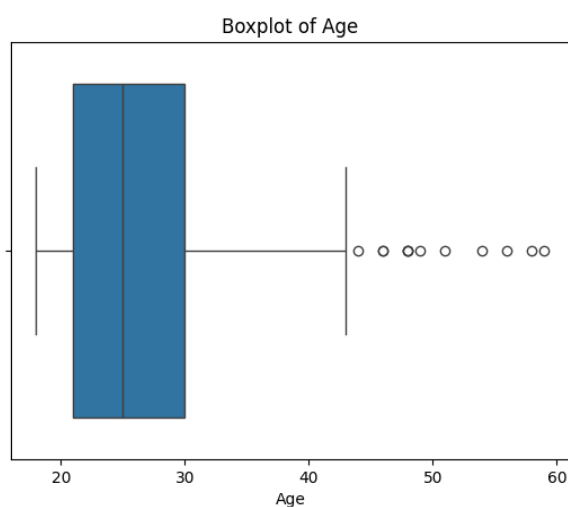


Figure 4.1(Box plot with outliers)

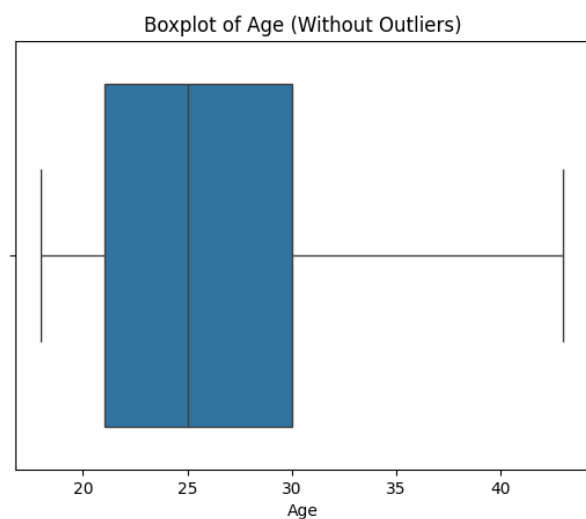


Figure 4.2(Box plot without outliers)

From fig 4.1 explains the dataset with outliers

From fig 4.2 explains the dataset without outliers

4.1.2 Descriptive Statistics

Descriptive statistics is the analysis of the data set, i.e., the data is summarized, and the data distribution is understood.

The statistical values that are used here are as follows:

- Mean
- Median
- Mode
- Standard Deviation
- Minimum and Maximum values

These statistical values help in understanding the trend of the following values:

- CGPA
- Sleep Duration
- Work/Study Hours
- Financial Stress

4.1.3 Data Visualization

Data visualization is used to represent data graphically, making it easier to understand patterns and relationships.

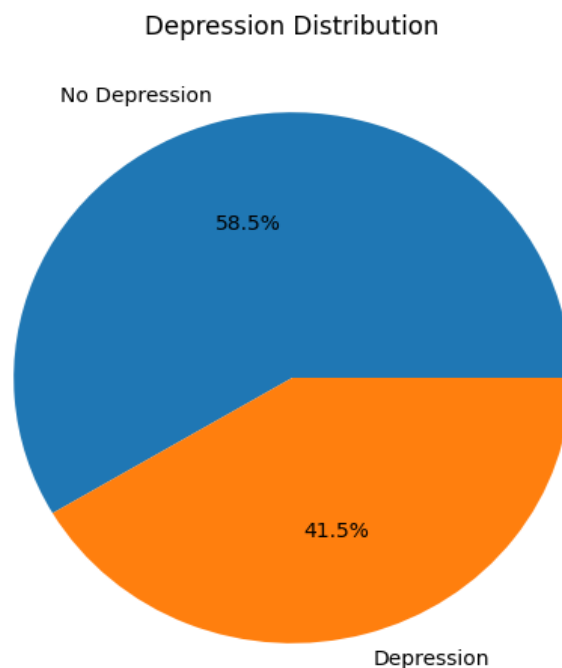


Figure 4.3 (Depression Distributor)

Fig 4.3 pie chart represents the ratio of students suffering from depression and those not suffering from depression. Each section represents the percentage contribution of each category.

The visualization is helpful in understanding whether the data is balanced or imbalanced.

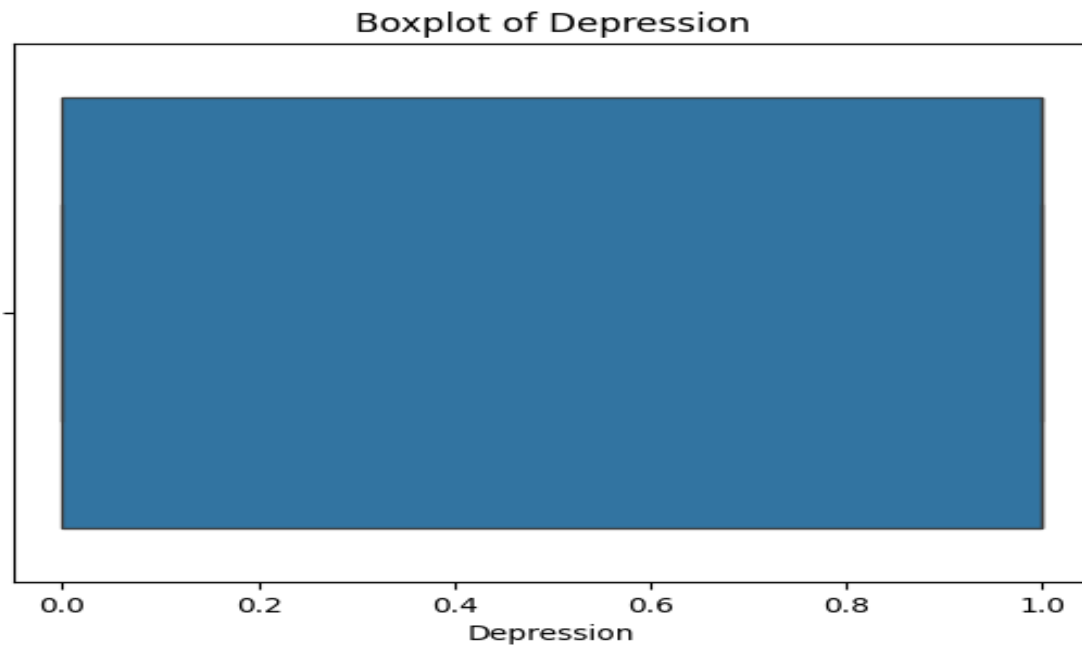


Figure 4.4 (Box Plot of Depression Count)

Fig 4.4 is a bar plot which is helpful in understanding the comparison of the number of students in each category.

From the visualization:

- One class - Depressed/Not Depressed - might dominate
- Understanding class imbalance

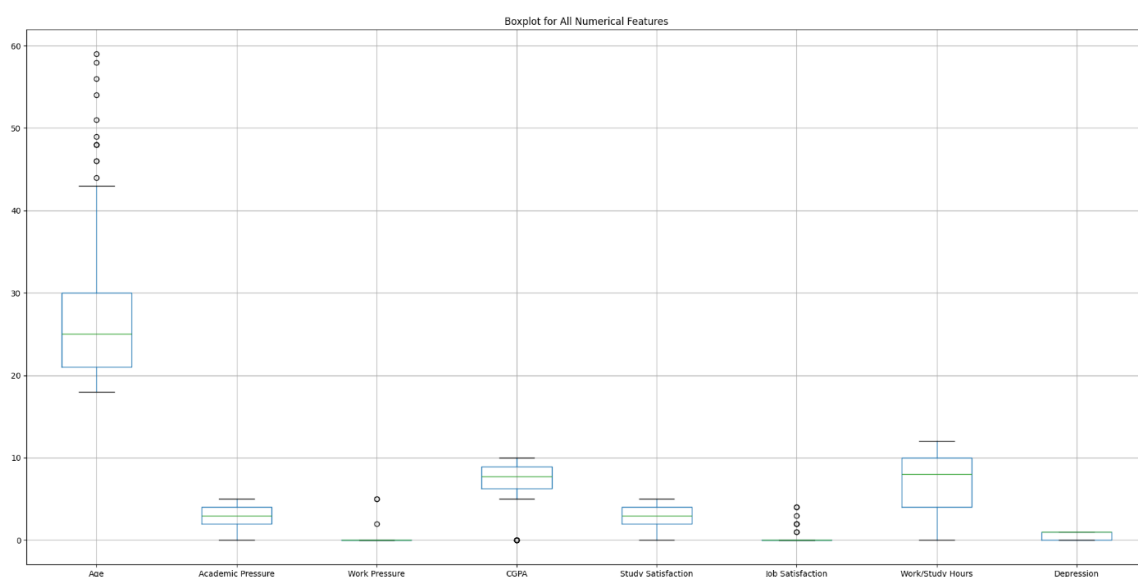


Figure 4.5(Box Plot for all Numerical Features)

Fig 4.5 shows a box plot is helpful in understanding:

- Median values
- Quartiles (Q1, Q3)
- Outliers

Fig. 4.5 plot is helpful in understanding features such as:

- CGPA
- Sleep Duration
- Work Hours

Outliers can affect the performance of models, so it is essential to understand them.

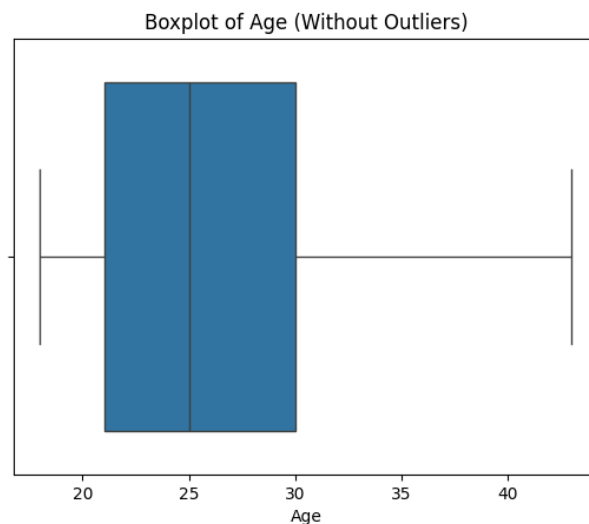


Figure 4.6(Box plot with outliers)

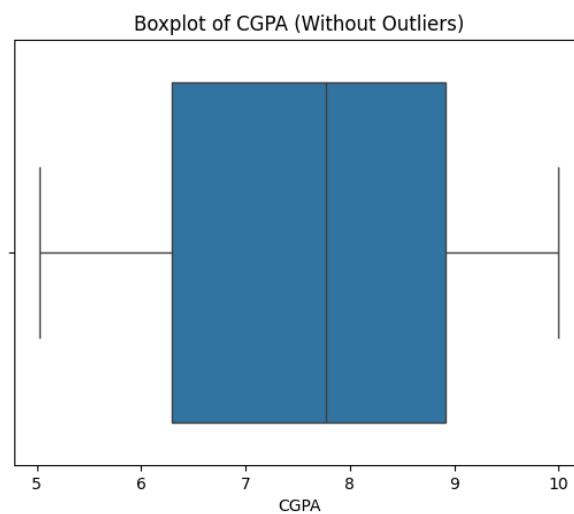


Figure 4.7(Box plot with outliers)

Fig. 4.6 and Fig.4.7 box plots help us in understanding the features after removing outliers.

4.1.4 Correlation & Relationships

The correlation matrix represents relationships between variables.

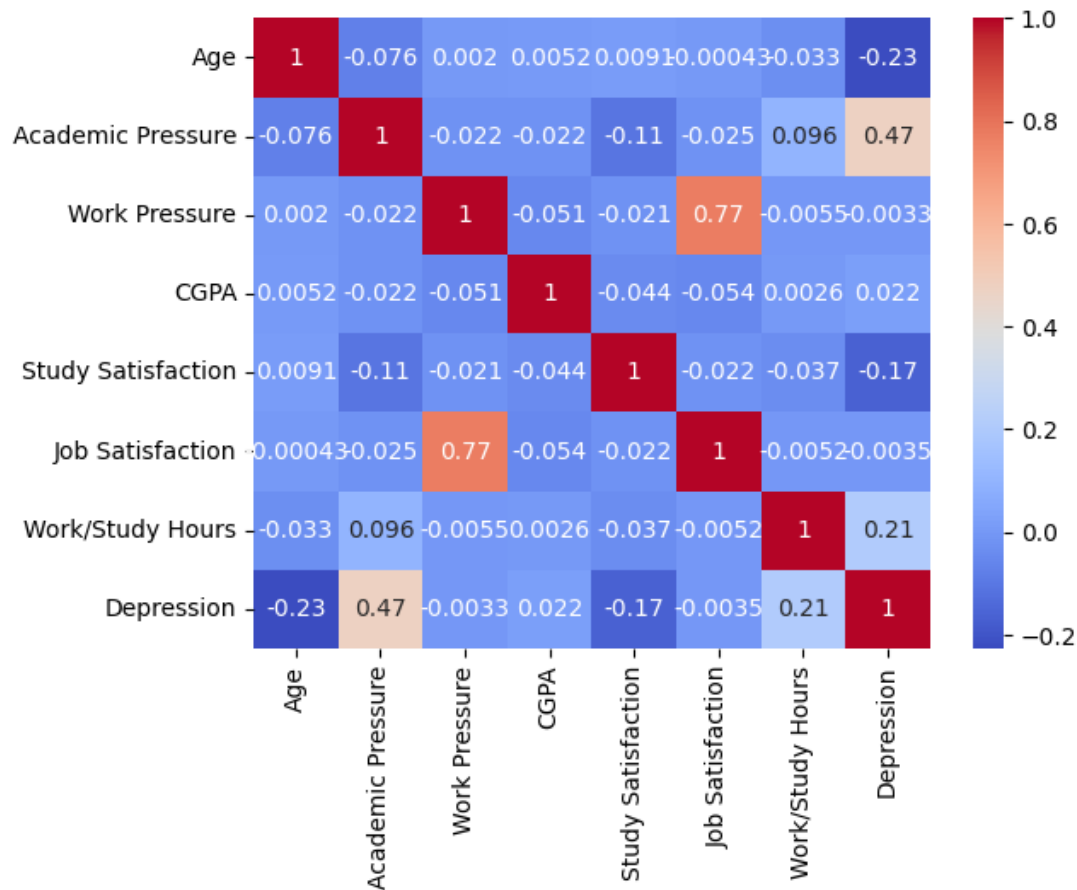


Figure 4.8 (Corelation Matrix)

Fig. 4.8 Shows us that:

- The values lie within the range -1 to +1
- If correlation is positive, then there is a direct relationship
- If correlation is negative, then there is an inverse relationship

Important relationships:

- Academic Pressure - Strong relationship with Depression
- Sleep Duration - Negative relationship
- Financial Stress - Positive relationship

Relationships:

- Depression and the Academic Pressure score have the strongest positive correlation, suggesting that this is the most important factor.
- There is a moderately positive correlation between work/study hours and depression, suggesting that students become more depressed the more hours they spend working
- There is a weak negative correlation between age and depression, meaning that students who are younger tend to be more depressed.
- There is no discernible relationship between depression and CGPA
- There is no discernible relationship between depression and work pressure or job satisfaction.
- In conclusion, depression is primarily caused by academic pressure, and depression can be lessened by satisfaction levels.

4.1.5 Data Segmentation

The given set of data is segmented according to:

- Gender
- Age Groups
- Academic Pressure
- Sleep Duration

From this segmentation, specific groups are identified that are more prone to depression.

4.1.6 Hypothesis Generation

Based on the findings in Exploratory Data Analysis, hypotheses are generated as follows:

- Students with high academic pressure are more prone to depression
- Low sleep duration is associated with depression
- Financial stress is associated with mental illness
- Low study satisfaction is associated with depression

4.1.7 Data Quality Assessment

The quality of the given set of data is assessed by considering:

- Completeness
- Consistency
- Accuracy

The dataset is found to be appropriate after preprocessing.

4.1.8 Types of Analysis Performed

1. Univariate Analysis

Analysis of individual variables

Example of Univariate Analysis

Analysis of distribution of CGPA

2. Bivariate Analysis

Analysis of relationship between two variables

Example of Bivariate Analysis

Analysis of relationship between sleep and depression

3. Multivariate Analysis

Analysis of multiple variables

Example of Multivariate Analysis

Analysis of relationship between academic pressure, sleep, and depression

4. Missing Data Analysis

Analysis of missing values

Use of imputation methods

5. Outlier Analysis

Analysis of extreme values

Use of box plot

4.1.9 Conclusion

Exploratory Data Analysis has played an important role in understanding the given dataset before proceeding with the application of machine learning models. From the analysis, it has been identified that there is an important relationship between different factors and depression. The results obtained from the analysis will be helpful in building an accurate and efficient predictive model.

4.2 Model Fitting

In the machine learning field, the method of model fitting is used as a technique of training the model on the data, with the goal of determining possible patterns that may exist between different input parameters of the data and the target variable. In the creation of a reliable prediction model, the importance of model fitting cannot be overstated. The ability of the

model, with the help of various input parameters, helps us understand the different kinds of relationships that exist with the different levels of depression.

4.2.1 Steps of Model Fitting

4.2.1.1 Selection of the Model: The selection of the appropriate algorithm, according to the problem domain, is the first step of the model fitting technique. In this problem, we select the models that can be used for classification, as the problem is about depression prediction.

4.2.1.2 Preparation of the Training Data: The second step of the model fitting technique is the preparation of the data that is needed for the model. In this problem, we divide the data into two sets: the first is used as the training data, while the second is used as the testing data. The most commonly used division is 70:30, i.e., 80:20.

4.2.1.3 Initialization of Parameters: The parameters of the model are given certain values before the actual training of the model commences. Then, these parameters are adjusted in such a manner that the performance of the model is optimized. This is done in regression models, for example, by adjusting the weights, and in classification models, for example, by adjusting the decision boundaries.

4.2.1.4 Iterative Optimization: The model learns to improve itself in terms of predictions by attempting to optimize an error or loss function. This is done by adjusting the parameters of the model in such a manner that the error between the actual and predicted values is minimized, using gradient descent.

4.2.1.5 Model Evaluation: The model is evaluated using various evaluation metrics after it has been trained. Some of these metrics include accuracy, precision, recall, F1 score, mean squared error, etc. This is done in order to determine the accuracy of the predictions made by the model, whether it is suffering from overfitting or underfitting, etc.

4.2.1.6 Hyperparameter Tuning: Hyperparameters such as learning rate, number of trees, depth of trees, etc. are tuned in order to improve the performance of the model. Grid search is used in this case.

4.2.1.7 Final Model Selection: The model with the best performance is selected and used in predicting new data that is unseen by the model.

4.2.2 Key Concepts in Model Fitting

4.2.2.1 Flexibility in Model Selection: The model used to fit the depression dataset is chosen from a variety of models. This increases the accuracy of the model in prediction.

4.2.2.2 Modifying Model Parameters: The parameters are adjusted during the training process in order to reduce the errors in the model's predictions and improve the model's ability to learn.

4.2.2.3 Overfitting and Underfitting: The overfitting model is one where the model is learning the "noise" in the data and performs badly on new data. Underfitting is where the model is not complex enough to learn patterns in the data. Therefore, we are looking at striking a balance in the model so that we can improve generalization.

4.2.2.4 Validation and Cross-Validation

The validation technique is used to check the performance of the model during training. Cross-validation, especially k-fold cross-validation, is used where we divide the data into different parts and ensure that we are testing the model on different parts of the data. This is done so that we don't overfit the model.

4.2.2.5 Final Assessment

The model is tested on new data (test data) to check the performance of the model in a real-world scenario. This is done so that we can ensure that the model is able to generalize and predict depression in a new case.

4.2.3 Models

4.2.3.1 Logistic Regression

Logistic Regression is one of the most popular statistical classification algorithms that is used for predicting binary outcomes such as whether a student is depressed or not. It works on the principle of using a sigmoid function that maps the predicted output between 0 and 1. In this study, Logistic Regression is used in understanding how different factors affect a person's depression. It is a simple and computationally fast algorithm that is easy to interpret.

Table 4.2.3.1

Train-Test Ratio	Accuracy
65-35	0.8473
60-40	0.8443
70-30	0.8479
75-25	0.8474
80-20	0.8471

- The above table 4.2.3.1 showing the accuracy of the Logistic Regression model with different train and test split ratios. It is observed that the maximum accuracy of 0.8479 is obtained with the 70:30 ratio.

4.2.3.2 K-Nearest Neighbors (KNN)

The K-Nearest Neighbors (KNN) algorithm is a non-parametric, instance-based learning algorithm that relies on classification based on the similarity of the features present in the data set. It relies on the determination of the classes present in the data set based on the majority classes found in the data set using the Euclidean distance algorithm. The KNN algorithm plays an important role in the identification of patterns present in the data set based on the characteristics of the students. However, the algorithm may not perform well depending on the selection of the 'K' parameter, and the algorithm may be expensive in terms of computational costs.

Table 4.2.3.2

Train-Test Ratio	Accuracy
65-35	0.7876
60-40	0.8035
70-30	0.7869
75-25	0.7880
80-20	0.7911

- The above table 4.2.3.2 shows the accuracy of the K-Nearest Neighbors (KNN) model with varying train-test split ratios. The maximum accuracy is 0.8035 with a 60:40 split ratio out of all experiments

4.2.3.3 Support Vector Classifier (SVC)

The Support Vector Classifier (SVC) algorithm is an efficient supervised learning algorithm that relies on the identification of the best hyperplane that separates the classes present in the data set. It works well with high-dimensional spaces, including linear as well as non-linear classification, with the kernel algorithm. The SVC algorithm plays an important role in the precise identification of the classes in the data set, identifying the depressed and non-depressed classes based on the features of the data set. It works well even when the data set is non-linearly separable.

Table 4.2.3.3

Train-Test Ratio	Accuracy
65-35	0.8413
60-40	0.8455
70-30	0.8421
75-25	0.8412
80-20	0.8424

- The above table 4.2.3.3 depicts the accuracy of the Support Vector Classifier (SVC) model along with various ratios of train-test splits. The highest accuracy of 0.8455 is achieved for the 60:40 split ratio for all experiments.

4.2.3.4 Decision Tree

Decision Tree is a classification tree-based model that divides the data set into smaller subsets based on the values of the features. Each node in the tree represents the decision rule, and the branches represent the outcomes of the decisions made. This helps in determining the factors that are the most influential in the development of depression, such as academic pressure or satisfaction with the course of study. However, this can cause overfitting if not checked.

Table 4.2.3.4

Train-Test Ratio	Accuracy
65-35	0.7781
60-40	0.7693
70-30	0.7751
75-25	0.7685
80-20	0.7735

- The above table 4.2.3.4 depicts the accuracy of the Decision Tree model along with different train-test split ratios is depicted in the above table. The highest accuracy of 0.7781 is achieved with a 65:35 split ratio in all experiments

4.2.3.5 Random Forest

Random Forest is a collection of algorithms that combine the outcomes of multiple decision trees in classification or regression tasks. It is used to reduce overfitting by averaging the outcomes of the trees created from the data set and the features. This is very effective in the prediction of depression as the complex relationships between the variables are well understood by the Random Forest algorithm, which makes it one of the best models.

Table 4.2.3.5

Train-Test Ratio	Accuracy
65-35	0.8424
60-40	0.8399
70-30	0.8388
75-25	0.8363
80-20	0.8392

- The above table 4.2.3.5 indicates the accuracy of the Random Forest model along with the different ratios of train-test splits. The highest accuracy of 0.8424 is achieved with the 65:35 split ratio in all experiments

4.2.3.6 Bagging (Bootstrap Aggregating)

Bagging is another ensemble method that uses a combination of models to improve the accuracy of predictions made by the system. This method works on the concept of reducing the variance of the data set, thus improving the stability of the system. In the proposed project, the

Bagging ensemble method is used to improve the efficiency of the system by reducing the possibility of overfitting, thus improving the accuracy of predictions made by the system.

Table 4.2.3.6

Train-Test Ratio	Accuracy
65-35	0.8229
60-40	0.8222
70-30	0.8228
75-25	0.8214
80-20	0.8241

- The above table 4.2.3.6 indicates the accuracy of the Bagging (Bootstrap Aggregating) model with varying train and test split ratios. The maximum accuracy of 0.8241 is achieved with the split ratio of 80:20.

4.2.3.7 AdaBoost

AdaBoost, or Adaptive Boosting, is another ensemble method used in machine learning for improving the accuracy of predictions made by the system. This method works on the concept of improving the accuracy of predictions made by the system, especially in classification problems. In the proposed project, AdaBoost is used for improving the accuracy of predictions made by the system in the classification of depression.

Table 4.2.3.7

Train-Test Ratio	Accuracy
65-35	0.8427
60-40	0.8416
70-30	0.8417
75-25	0.8404
80-20	0.8424

- The above table 4.2.3.7 indicates the accuracy of the AdaBoost algorithm along with various ratios of train-test data. The highest accuracy of 0.8427 is achieved for the 65:35 split ratio in all experiments.

4.2.3.8 Naive Bayes

Naive Bayes is a probabilistic classifier that is developed on the basis of Bayes' theorem. It is computationally efficient as it assumes that the features are independent of each other, which makes the calculations simpler. Despite the simplicity of the algorithm, it is effective in solving many classification problems, especially the prediction of depression.

Table 4.2.3.8

Train-Test Ratio	Accuracy
65-35	0.4233
60-40	0.4425
70-30	0.4205
75-25	0.4239
80-20	0.4208

- The above table 4.2.3.8 is representing the accuracy of Naive Bayes with different ratios of train-test split. The maximum accuracy of 0.4425 is achieved with 60:40 split ratio out of all experiments.

4.2.3.9 Gradient Boosting

Gradient Boosting is an advanced form of the ensemble method, in which the models are developed sequentially, with the objective of correcting the mistakes of the previous model. Gradient descent is used to minimize the loss function, making the algorithm highly effective in solving many real-world problems, especially the prediction of depression.

Table 4.2.3.9

Train-Test Ratio	Accuracy
65-35	0.8459
60-40	0.8453
70-30	0.8469
75-25	0.8471
80-20	0.8455

- The above table 4.2.3.9 indicates the accuracy of the Gradient Boosting model along with different ratios of train-test data. The highest accuracy of 0.8471 is achieved with the 75:25 split ratio in all experiments.

4.2.3.10 Multi-Layer Perceptron (MLP)

An artificial neural network is called a Multi-Layer Perceptron (MLP), which has one input layer, one hidden layer, and one output layer. The MLP has the ability to learn patterns using activation functions and backpropagation on the patterns that it has seen so far. In terms of predicting depression, MLP is significant because it makes features linearly non-related, thereby providing the model with a lot of predictive power.

Table 4.2.3.10

Train-Test Ratio	Accuracy
65-35	0.8075
60-40	0.8268
70-30	0.8049
75-25	0.8158
80-20	0.8179

- The above table 4.2.3.10 shows the accuracy of the Multi-Layer Perceptron (MLP) model with different train and test split ratios. The highest accuracy of 0.8268 is obtained for the 60-40 split ratio among all experiments.

4.2.3.11 Stochastic Gradient Descent (SGD)

Stochastic Gradient Descent (SGD) is a technique that is used to optimize a machine learning model so that it learns faster. The optimization algorithm is significant when one is dealing with large datasets. SGD is significant in this project because it makes the model better and reduces the time taken by the model to learn

Table 4.2.3.11

Train-Test Ratio	Accuracy
65-35	0.8420
60-40	0.8360
70-30	0.8490
75-25	0.8328
80-20	0.7386

- The above table 4.2.3.11 indicates the accuracy of the Stochastic Gradient Descent (SGD) model along with various ratios of train-test splits. The highest accuracy of 0.8490 is achieved for the 70:30 split ratio in all the experiments.

4.2.3.12 LightGBM (Light Gradient Boosting Machine)

LightGBM is an efficient gradient boosting framework based on tree learning. It is used for big data and has good performance in both fast training and high accuracy. In the prediction of depression, LightGBM has good performance in feature interaction and low costs of computation.

Table 4.2.3.12

Train-Test Ratio	Accuracy
65-35	0.8453
60-40	0.8441
70-30	0.8484
75-25	0.8458
80-20	0.8473

- The above table 4.2.3.12 shows the accuracy of the LightGBM model with different train and test split ratios. The highest accuracy of 0.8484 is obtained using the 70-30 split ratio for all experiments.

4.2.3.13 CatBoost

CatBoost is a gradient boosting algorithm. CatBoost avoids overfitting and reduces the need for data preprocessing using ordered boosting. In this study, CatBoost is used to enhance the accuracy of the model and handle categorical features.

Table 4.2.3.13

Train-Test Ratio	Accuracy
65-35	0.8458
60-40	0.8439
70-30	0.8473
75-25	0.8454
80-20	0.8507

- The above table 4.2.3.13 indicates the accuracy of the CatBoost model with different ratios of train and test sets. The highest accuracy of 0.8507 is achieved with the split ratio of 80:20.

4.2.3.14 XGBoost

XGBoost stands for Extreme Gradient Boosting and is one of the best boosting algorithms used today due to its speed and performance. It also includes various regularizations that avoid overfitting and provide the best results. Using the XGBoost algorithm in the prediction of depression gives the best accuracy as the relationships between the variables are complex.

Table 4.2.3.14

Train-Test Ratio	Accuracy
65-35	0.8371
60-40	0.8379
70-30	0.8400
75-25	0.8402
80-20	0.8397

- The table above table 4.2.3.14 indicates the accuracy of the XGBoost model with various train-test split ratios. The maximum accuracy of 0.8402 is achieved when using a 75:25 ratio out of all the experiments.

In the above study, various machine learning algorithms are implemented and compared to effectively predict depression. The algorithms like Random Forest, Gradient Boosting, XGBoost, and CatBoost are performing well as they are able to handle complex relationships between variables and avoid overfitting. Choosing the best algorithm, accurate and reliable prediction can detect depression at the right time.

4.3 Forecasting

The concept of forecasting can be defined as the prediction of future outcomes based on the patterns learned from the past data. Forecasting is employed as the key concept in this project to predict the future outcomes of whether the student is likely to develop depression or not based on the input features.

Machine learning models are trained on the past data of the students to learn the patterns and relationships between the input features and the depression level of the students. The trained models are then employed to make predictions on the future data or unseen data. The forecasting outcomes are employed to identify the students at high risk of developing depression.

Unlike the conventional forecasting techniques employed in the time series analysis, this study employs classification-based forecasting techniques to predict the outcomes as follows:

- Depressed (High Risk)
- Not Depressed (Low Risk)

The accuracy of the forecasting outcomes depends on the accuracy of the data and the performance of the employed machine learning models.

Table 4.8

S.No	Model	Accuracy	Prediction Level
1	Logistic Regression	0.8479	High Accuracy
2	KNN	0.8035	Moderate Accuracy
3	SVC	0.8455	High Accuracy
4	Decision Tree	0.7781	Moderate Accuracy
5	Random Forest	0.8424	High Accuracy
6	Bagging	0.8241	High Accuracy
7	AdaBoost	0.8427	High Accuracy
8	Naive Bayes	0.4425	Low Accuracy
9	Gradient Boosting	0.8471	High Accuracy
10	MLP	0.8268	High Accuracy
11	SGD	0.8490	High Accuracy
12	LightGBM	0.8484	High Accuracy
13	CatBoost	0.8507	Very High Accuracy
14	XGBoost	0.8402	High Accuracy

From the above table 4.8:

- If the accuracy of the model is above 0.84, it means that the model has high forecasting ability
- CatBoost, with an accuracy of 0.8507, had the highest accuracy, meaning it was the best model for forecasting
- SGD and LightGBM had equally impressive performances
- Naive Bayes had the least accuracy, implying that it had the poorest forecasting ability
- Ensemble and boosting models are the best in forecasting

From the forecasting analysis, it is clear that ensemble models such as CatBoost, LightGBM, and Gradient Boosting are the best in terms of forecasting accuracy for the detection of depression in students. These models can be applied in real-world scenarios for the identification of students with depression.

Chapter-5

5.1 Results

In this research work, various models were implemented using machine learning algorithms that were used to predict depression among students based on various features such as academic pressure, CGPA, sleeping hours, work/study hours, financial problems, and satisfaction levels. The performance of each model was also tested using various ratios.

The performance of each model was also tested using accuracy, which is a measure of how well each individual model classified students as depressed or not depressed. The results obtained using each individual model showed that all models were able to achieve good accuracy. This proved that the machine learning model had the ability to accurately predict the mental health conditions of students.

The models that showed good performance were Logistic Regression with a value of 0.8479, Support Vector Classifier with a value of 0.8455, Gradient Boosting with a value of 0.8471, and LightGBM with a value of 0.8484.

The good performance shown by the models proved that the linear models as well as the ensemble models were good at predicting the depression of students. The other models, such as Random Forest with a value of 0.8424, Bagging with a value of 0.8241, and AdaBoost with a value of 0.8427, showed good performance as they were able to prevent overfitting.

The Multi-Layer Perceptron (0.8268) showed the ability of neural networks to handle non-linear relationships between features and depression. Stochastic Gradient Descent (0.8490) also showed a high accuracy, thereby ensuring that the model learns efficiently on the given data.

However, models such as K-Nearest Neighbors (0.8035) and Decision Tree (0.7781) showed a comparatively low accuracy, which might be because these models are sensitive to the data distribution as well as parameters. Naive Bayes showed the least accuracy (0.4425), which is because this model is based on the assumption that features are independent, but in this case, features are correlated.

From the results, it is clear that ensemble models as well as boosting algorithms perform better when compared to traditional models. These models are able to handle complex relationships between features as well as non-linear relationships that are present in the given data. Academic pressure, hours spent on work/study, as well as study satisfaction, are significant factors that influence depression, and models that are able to learn this effectively tend to produce better accuracy.

Another notable observation is that the ratio of splitting the train and test sets has an impact on the performance of the models. The best accuracy of each model was obtained using different ratios of splitting the train and test sets. This shows that the performance of models is affected by the splitting of the train and test sets.

Overall, based on the results obtained from the experiments, it is clear that machine learning models can be used for predicting depression among students.

Out of all the models implemented during this research, CatBoost was identified as the best model for depression prediction, with an accuracy of 0.8507 using an 80-20 ratio for splitting the train and test sets. This is because CatBoost can work well with categorical features and avoid overfitting.

Chapter-6

6.1 Comparison of results

Table 6.1

Model	Highest Accuracy	Train-Test Split
CatBoost	0.8507	80:20
Stochastic Gradient Descent (SGD)	0.8490	70:30
LightGBM	0.8484	70:30
Logistic Regression	0.8479	70:30
Gradient Boosting	0.8471	75:25
Support Vector Classifier (SVC)	0.8455	60:40
AdaBoost	0.8427	65:35
Random Forest	0.8424	65:35
XGBoost	0.8402	75:25
Multi-Layer Perceptron (MLP)	0.8268	60:40
Bagging	0.8241	80:20
K-Nearest Neighbors (KNN)	0.8035	60:40
Decision Tree	0.7781	65:35
Naive Bayes	0.4425	60:40

The above comparison table 6.1 represents the performance of different machine learning models used for depression prediction based on the best accuracy achieved by the models and the split ratios used for training and testing the models. Among the models, the CatBoost algorithm achieved the best accuracy of 0.8507 with a split ratio of 80:20 for training and testing the model. This suggests that the CatBoost algorithm is the best among the models used for depression prediction. Other models such as Stochastic Gradient Descent achieved an accuracy of 0.8490, LightGBM achieved an accuracy of 0.8484, and Logistic Regression achieved an accuracy of 0.8479. All these models are performing well with the dataset, which suggests that the models are effective in handling the dataset. Ensemble and boosting algorithms such as Gradient Boosting, AdaBoost, Random Forest, and XGBoost achieved high accuracy with the dataset due to the nature of the algorithms. The models such as MLP, Bagging, and KNN achieved moderate accuracy with the dataset, whereas the Decision Tree algorithm achieved less accuracy with the dataset. The Naive Bayes algorithm achieved the least accuracy with the dataset.

6.2 Discussion

The objective of this study was to examine the student mental health indicators and predict the depression level using various machine learning algorithms. The data set contained various features, i.e., academic pressure, CGPA, sleep duration, work/study hours, financial pressure, study satisfaction, etc. Using various machine learning algorithms, the study was aimed at determining the best model that could accurately predict the depression level.

The results obtained from the experimental analysis show that the machine learning algorithms performed well in predicting the depression level, with accuracy ranging from moderate to high levels. The comparison of the machine learning algorithms shows that the ensemble algorithms performed better in predicting the depression level, as they are capable of dealing with complex relationships between the features as well as the non-linear data.

Among all the models, CatBoost has the highest accuracy of 0.8507 with an 80:20 ratio. Therefore, CatBoost is the most accurate model. This is because CatBoost is most efficient in handling categorical features and reducing overfitting using ordered boosting. In addition, Stochastic Gradient Descent (0.8490) and LightGBM (0.8484) have also demonstrated impressive accuracy. This is an indication that Stochastic Gradient Descent and LightGBM are most efficient in handling large data and optimizing model parameters. Other models such as Logistic Regression (0.8479) and Support Vector Classifier (0.8455) have also demonstrated impressive accuracy. This is an indication that linear and margin-based classifiers are most efficient in modeling the relationship between features and depression. In addition, other ensemble methods such as Gradient Boosting (0.8471), AdaBoost (0.8427), and Random Forest (0.8424) have also demonstrated impressive accuracy. This is an indication that ensemble methods are most efficient in reducing overfitting. Moderate accuracy has been demonstrated in other models such as Multi-Layer Perceptron (0.8268), Bagging (0.8241), and K-Nearest Neighbors (0.8035). These are most efficient in modeling patterns. However, Decision Trees (0.7781) have demonstrated low accuracy. This is because Decision Trees are most likely to overfit and are not general.

The accuracy of the Naive Bayes model was the lowest among all models, at 0.4425. This is because this model is based on independent features, and this does not hold for this dataset. For example, academic pressure, financial stress, and satisfaction levels while studying are all interconnected.

It is also important to note from this research that the train-test split ratio plays an important role in determining how well a model will perform. For example, the best accuracy for CatBoost was obtained using an 80-20 ratio, while for other models such as SGD and LightGBM, an 70-30 ratio was used.

It is also clear from these results that certain features such as academic pressure, work and study hours, and satisfaction levels while studying play an important role in determining the levels of depression. This is because these features have shown strong correlation with depression levels.

It is clear from this research that machine learning models can be used for the early detection of depression levels among students.

6.2.1 Conclusion

This study was successful in applying various machine learning algorithms to predict depression among students based on different demographic factors, academic factors, lifestyle factors, and psychological factors.

The analysis and the results of the experiment confirm the efficiency of using machine learning as a tool to identify mental health risks.

The results obtained in the experiment show that the better algorithms, such as ensemble algorithms and boosting algorithms, are better than the basic ones. It was observed that the best algorithm among all the algorithms used in this research was the CatBoost algorithm, which has an accuracy of 0.8507 using the 80:20 split ratio.

The results of the research emphasize the significance of key factors such as academic pressure, work/study hours, and satisfaction levels in the prediction of student mental health. The model can successfully predict the mental health of students by analyzing these key factors.

The prediction system designed by this research can be implemented in various applications, such as in the fields of education, counselling systems, and mental health monitoring systems.

In conclusion, the research clearly indicates the effectiveness, reliability, and efficiency of the machine learning approach in the prediction of depression. The results of the research can act as a foundation for future research in the field of mental health analysis.

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Appendix

Importing libraries

```
# importing libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
```

Importing the data

```
#loading the data set
df=pd.read_csv('/content/student_depression_dataset (1).csv')
```

Shape (no. of rows and columns)

```
df.shape
```

Describing the data

```
df.describe()
```

Reading the first 5 rows

```
df.head(5)
```

Reading the last 5 rows

```
df.tail(5)
```

Names of the columns

```
df.columns
```

Checking for duplicate values

```
df.duplicated()
```

Checking for Null values

```
df.isnull().sum()
```

Dropping the columns

```
df = df.drop(columns=['id'])
```

```
x = df.drop(columns=['Depression'])
```

Again, checking for the columns

```
df.columns
```

Data Visualisation:

Box plot before outliers

```
sns.boxplot(x='Age', data=df)  
plt.title('Boxplot of Age')  
plt.show()
```

Box plot after outliers

```
q1 = df['Age'].quantile(0.25)
q3 = df['Age'].quantile(0.75)
iqr = q3 - q1
print("IQR for Age", iqr)
upperrange = q3 + 1.5 * iqr
bottomrange = q1 - 1.5 * iqr
df_clean = df[(df['Age'] > bottomrange) & (df['Age'] < upperrange)]
sns.boxplot(x='Age', data=df_clean)
plt.title('Boxplot of Age (Without Outliers)')
plt.show()
```

Boxplot for All Numerical Columns

```
#Boxplot for All Numerical Columns
df.boxplot(figsize=(25, 12))
plt.title('Boxplot for All Numerical Features')
plt.show()
```

Histograms for All Columns

```

# Histograms for All Columns
# Draw histograms for each column in the dataset

# Select only numeric columns for plotting
numeric_columns = df.select_dtypes(include=['number']).columns
num_columns = len(numeric_columns)
num_rows = (num_columns + 1) // 2 # Calculate the number of rows for subplots

fig, axes = plt.subplots(nrows=num_rows, ncols=2, figsize=(12, num_rows * 4))

# Flatten axes if multiple rows are used
axes = axes.flatten()

for i, column in enumerate(numeric_columns):
    df[column].hist(ax=axes[i], bins=20) # Adjust bins as needed
    axes[i].set_title(f"{column}")
    axes[i].set_xlabel(column)
    axes[i].set_ylabel("Frequency")

# If there's an odd number of columns, remove the empty subplot
if num_columns % 2 != 0:
    fig.delaxes(axes[num_columns])

plt.tight_layout() # Adjust layout for spacing between subplots
plt.show()

```

Pair plots for all columns

```

import seaborn as sns
import matplotlib.pyplot as plt

selected_cols = [
    'Age',
    'CGPA',
    'Work/Study Hours',
    'Academic Pressure',
    'Financial Stress'
]

sns.pairplot(df[selected_cols + ['Depression']], hue='Depression')
plt.show()

```

Count plots for all columns

```
cat_columns = [
    'Gender',
    'City',
    'Profession',
    'Sleep Duration',
    'Dietary Habits',
    'Degree',
    'Have you ever had suicidal thoughts ?',
    'Family History of Mental Illness'
]

num_cols = len(cat_columns)
num_rows = (num_cols + 1) // 2

fig, axes = plt.subplots(nrows=num_rows, ncols=2, figsize=(14, num_rows * 4))
axes = axes.flatten()

for i, col in enumerate(cat_columns):
    sns.countplot(x=col, data=df, ax=axes[i])
    axes[i].set_title(f"Count Plot for {col}")
    axes[i].set_xlabel(col)
    axes[i].set_ylabel("Count")
    axes[i].tick_params(axis='x', rotation=45)

# Remove extra empty subplot if odd number of plots
if num_cols % 2 != 0:
    fig.delaxes(axes[-1])

plt.tight_layout()
plt.show()
```

Pie chart

```
counts = df['Depression'].value_counts()

labels = ['No Depression', 'Depression']

plt.figure(figsize=(6,6))
plt.pie(counts, labels=labels, autopct='%1.1f%%')
plt.title('Depression Distribution')
plt.show()
```

Subplots for all columns

```
# Define features and target
features = X.columns
target = 'Depression'

# Number of columns for subplot layout
num_cols = 3
num_features = len(features)
num_rows = -(-num_features // num_cols) # ceiling division

# Create subplots
fig, axes = plt.subplots(num_rows, num_cols, figsize=(15, num_rows*4))
axes = axes.flatten()

# Plot scatter plots of each feature vs target
for i, feature in enumerate(features):
    axes[i].scatter(X[feature], df[target], alpha=0.5)
    axes[i].set_xlabel(feature)
    axes[i].set_ylabel(target)
    axes[i].set_title(f'{feature} vs {target}')

# Remove unused subplots
for j in range(i+1, len(axes)):
    fig.delaxes(axes[j])

plt.tight_layout()
plt.show()
```

Bar plots

```
sns.barplot(x='Gender', y='Depression', hue='Family History of Mental Illness', data=df)
plt.title('Depression by Gender and Family History of Mental Illness')
plt.ylabel('Mean Depression (0 = No, 1 = Yes)')
plt.show()
```

Corelation matrix

```
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
```

Model Fitting:

Logistic regression

```
from sklearn.linear_model import LogisticRegression
```

```
model=LogisticRegression()
```

```
model.fit(x_train,y_train)
```

```
y_pred = model.predict(x_test)  
y_pred
```

```
y_pred = model.predict(x_test)  
y_prob = model.predict_proba(x_test)[:,-1]
```

```
Logistic = pd.DataFrame({'Predicted':y_pred,'Actual':y_test})  
Logistic
```

```
from sklearn.metrics import accuracy_score  
accuracy_score(y_test, y_pred)
```

```
from sklearn.metrics import classification_report  
print(classification_report(y_test,y_pred))
```

```
from sklearn.metrics import confusion_matrix  
  
cm = confusion_matrix(y_test, y_pred)  
print("Confusion Matrix:\n", cm)
```

```

sns.heatmap(
    cm,
    annot=True,
    fmt="d",
    cmap="Blues",
    xticklabels=["No Depression", "Depression"],
    yticklabels=["No Depression", "Depression"]
)

plt.title('Confusion Matrix - Depression Prediction')
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()

```

```

from sklearn.metrics import roc_curve, auc

fpr, tpr, thresholds = roc_curve(y_test, y_prob)
roc_auc = auc(fpr, tpr)

plt.figure(figsize=(6,4))
plt.plot(fpr, tpr, label='AUC = %0.2f' % roc_auc)
plt.plot([0,1], [0,1], linestyle='--')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('ROC Curve for Depression Prediction')
plt.legend()
plt.show()

print("ROC-AUC Score:", roc_auc)

```

KNN Classifier

```

from sklearn.neighbors import KNeighborsClassifier

model=KNeighborsClassifier(n_neighbors=20)

model.fit(x_train, y_train)

```

Svm Classifier

```
from sklearn.svm import SVC

model = SVC(kernel='linear')

model.fit(x_train, y_train)
```

Desion Tree Classifier

```
from sklearn.tree import DecisionTreeClassifier

model = DecisionTreeClassifier()

model.fit(x_train, y_train)
```

Random forest

```
from sklearn.ensemble import RandomForestClassifier

model = RandomForestClassifier()

model.fit(x_train, y_train)
```

Bagging

```
from sklearn.ensemble import BaggingClassifier

model = BaggingClassifier()

model.fit(x_train, y_train)
```

Ada Boost

```
from sklearn.ensemble import AdaBoostClassifier

model = AdaBoostClassifier()

model.fit(x_train, y_train)
```

Gradient Boost

```
from sklearn.ensemble import GradientBoostingClassifier

model = GradientBoostingClassifier()

model.fit(x_train, y_train)
```

XG Boost

```
from xgboost import XGBClassifier

model = XGBClassifier()

model.fit(x_train, y_train)
```

Navie Bayes Classifier

```
from sklearn.naive_bayes import GaussianNB

model = GaussianNB()

model.fit(x_train, y_train)
```

Multi-Layer Perception

```
from sklearn.neural_network import MLPClassifier

model = MLPClassifier()

model.fit(x_train, y_train)
```

Stochastic Gradient Descent (SGD) Classifier

```
from sklearn.linear_model import SGDClassifier

model = SGDClassifier()

model.fit(x_train, y_train)
```

CatBoost Classifier

```
!pip install catboost
```

```
from catboost import CatBoostClassifier

model = CatBoostClassifier()

model.fit(x_train, y_train)
```

LightGBM (Light Gradient Boosting Machine)

```
!pip install lightgbm
```

```
from lightgbm import LGBMClassifier

model = LGBMClassifier()

model.fit(x_train, y_train)
```

Best model(Based on accuracy):

```
# Find the row with the highest Accuracy
best_model_row = results_df.loc[results_df["Accuracy"].idxmax()]

# Display the best model's results
print("\n==== Best Model Performance (Based on Accuracy) ==== \n")
display(best_model_row)
```

Ann (Artificail Neural Network)

```
import tensorflow as tf

tf.random.set_seed(42)

# STEP1: Creating the model

model= tf.keras.Sequential([
    tf.keras.layers.Dense(10),
    tf.keras.layers.Dense(5),
    tf.keras.layers.Dense(1)
])

# STEP2: Compiling the model # optimizer can be SGD, Adam

model.compile(loss= tf.keras.losses.mae,
              optimizer= tf.keras.optimizers.Adam(), #SGD
              metrics= ["mae"])

# STEP3: Fit the model

history= model.fit(x_train, y_train, epochs= 10, verbose=1)
```

```
model.evaluate(x_test, y_test)
```

```
model.summary();
```

```
pd.DataFrame(history.history).plot()  
plt.ylabel("loss")  
plt.xlabel("epochs")
```

```
# Convert training history to DataFrame  
history_df = pd.DataFrame(history.history)  
  
# Plot all metrics  
history_df.plot(figsize=(10, 6))  
  
# Add title and labels  
plt.title("Training Metrics Over Epochs")  
plt.xlabel("Epochs")  
plt.ylabel("Value")  
plt.grid(True)  
plt.show()
```

Different Train test splits

```
from sklearn.model_selection import train_test_split  
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, roc_auc_score  
from sklearn.linear_model import LogisticRegression, SGDClassifier  
from sklearn.neighbors import KNeighborsClassifier  
from sklearn.svm import SVC  
from sklearn.tree import DecisionTreeClassifier  
from sklearn.ensemble import RandomForestClassifier, BaggingClassifier, AdaBoostClassifier, GradientBoostingClassifier  
from sklearn.naive_bayes import GaussianNB  
from sklearn.neural_network import MLPClassifier  
from lightgbm import LGBMClassifier  
import xgboost as xgb  
import pandas as pd
```

```
# ===== Models =====
def get_models():
    return {
        "Logistic Regression": LogisticRegression(max_iter=1000),
        "KNN": KNeighborsClassifier(),
        "SVC": SVC(probability=True),
        "Decision Tree": DecisionTreeClassifier(random_state=42),
        "Random Forest": RandomForestClassifier(random_state=42),
        "Bagging": BaggingClassifier(random_state=42),
        "AdaBoost": AdaBoostClassifier(random_state=42),
        "Gradient Boosting": GradientBoostingClassifier(random_state=42),
        "XGBoost": xgb.XGBClassifier(eval_metric='logloss', random_state=42),
        "Naive Bayes": GaussianNB(),
        "MLP": MLPClassifier(hidden_layer_sizes=(64,32), max_iter=500, random_state=42),
        "SGD": SGDClassifier(loss='log_loss', random_state=42),
        "LightGBM": LGBMClassifier(random_state=42),
        "CatBoost": CatBoostClassifier(verbose=0, random_state=42)
    }
```

```
# ===== Evaluation =====
def evaluate_model(y_true, y_pred, y_prob):
    return {
        "Accuracy": accuracy_score(y_true, y_pred),
        "Precision": precision_score(y_true, y_pred),
        "Recall": recall_score(y_true, y_pred),
        "F1-Score": f1_score(y_true, y_pred),
        "ROC-AUC": roc_auc_score(y_true, y_prob)
    }
```

```
# ===== Run for different splits =====
splits = [0.20, 0.25, 0.30, 0.35]
all_results = []

for test_size in splits:
    X_train, X_test, y_train, y_test = train_test_split(
        x, y, test_size=test_size, random_state=42
    )

    models = get_models()

    for name, model in models.items():
        model.fit(X_train, y_train)
        y_pred = model.predict(X_test)
        # predict_proba is not available for all models (e.g., SGDClassifier with certain loss functions)
        if hasattr(model, "predict_proba"):
            y_prob = model.predict_proba(X_test)[:, 1]
        else:
            # For models without predict_proba, use dummy probabilities or handle as appropriate
            y_prob = [0.5] * len(y_test) # Placeholder

        metrics = evaluate_model(y_test, y_pred, y_prob)
        metrics["Model"] = name
        metrics["Train-Test Split"] = f"{int((1-test_size)*100)}-{int(test_size*100)}"
        all_results.append(metrics)

# Convert to DataFrame
results_df = pd.DataFrame(all_results)
results_df
```

```
# Convert results into DataFrame (run this once after your training loop)
results_df = pd.DataFrame(all_results)

# Show results separately for each test split
for split in results_df["Train-Test Split"].unique():
    split_df = results_df[results_df["Train-Test Split"] == split].drop(columns=["Train-Test Split"])
    print(f"\n==== Results for Split {split} ==== \n")
    display(split_df)

    # Optional: quick bar chart for R2 comparison
    split_df.plot(x="Model", y="ROC-AUC", kind="bar", title=f"ROC-AUC Scores ({split} Split)")
    plt.show()
```

ANN Variants

Model 1 - Deep ANN with ELU + Adam

```
# Model 1 - Deep ANN with ELU + Adam (Classification)

model1 = tf.keras.Sequential([
    tf.keras.layers.Dense(64, activation='elu', input_shape=(X_train.shape[1],)),
    tf.keras.layers.Dense(32, activation='elu'),
    tf.keras.layers.Dense(16, activation='elu'),
    tf.keras.layers.Dense(8, activation='elu'),
    tf.keras.layers.Dense(1, activation='sigmoid') # classification output
])

# Compile
model1.compile(
    optimizer=tf.keras.optimizers.Adam(learning_rate=0.0008),
    loss='binary_crossentropy',
    metrics=['accuracy']
)

# Train
history1 = model1.fit(
    X_train, y_train,
    validation_split=0.2,
    epochs=100,
    batch_size=32,
    verbose=0
)

# Predict probabilities
y_prob1 = model1.predict(X_test).flatten()

# Convert to class labels
y_pred1 = (y_prob1 >= 0.5).astype(int)

# Evaluate
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, roc_auc_score

metrics1 = {
    "Accuracy": accuracy_score(y_test, y_pred1),
    "Precision": precision_score(y_test, y_pred1),
    "Recall": recall_score(y_test, y_pred1),
    "F1-Score": f1_score(y_test, y_pred1),
    "ROC-AUC": roc_auc_score(y_test, y_prob1)
}

metrics1["Model1"] = "ANN - Deep ELU Adam"
metrics1
```

Model 2 - Simple ANN with tanh + RMSprop

```
# Model 2 - Simple ANN with tanh + RMSprop (Classification)

model2 = tf.keras.Sequential([
    tf.keras.layers.Dense(32, activation='tanh', input_shape=(X_train.shape[1],)),
    tf.keras.layers.Dense(16, activation='tanh'),
    tf.keras.layers.Dense(1, activation='sigmoid') # classification output
])

# Compile
model2.compile(
    optimizer=tf.keras.optimizers.RMSprop(learning_rate=0.001),
    loss='binary_crossentropy',
    metrics=['accuracy']
)

# Train
history2 = model2.fit(
    X_train, y_train,
    validation_split=0.2,
    epochs=100,
    batch_size=32,
    verbose=0
)

# Predict probabilities
y_prob2 = model2.predict(X_test).flatten()

# Convert to class labels
y_pred2 = (y_prob2 >= 0.5).astype(int)

# Evaluate
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, roc_auc_score

metrics2 = {
    "Accuracy": accuracy_score(y_test, y_pred2),
    "Precision": precision_score(y_test, y_pred2),
    "Recall": recall_score(y_test, y_pred2),
    "F1-Score": f1_score(y_test, y_pred2),
    "ROC-AUC": roc_auc_score(y_test, y_prob2)
}

metrics2["Model"] = "ANN - Simple tanh RMSprop"
metrics2
```

Model 3 - Simple ANN with SELU + Nadam

```
# Model 3 - Simple ANN with SELU + Nadam (Classification)

model3 = tf.keras.Sequential([
    tf.keras.layers.Dense(32, activation='selu', input_shape=(X_train.shape[1],)),
    tf.keras.layers.Dense(16, activation='selu'),
    tf.keras.layers.Dense(1, activation='sigmoid') # classification output
])

# Compile
model3.compile(
    optimizer=tf.keras.optimizers.Nadam(learning_rate=0.001),
    loss='binary_crossentropy',
    metrics=['accuracy']
)

# Train
history3 = model3.fit(
    X_train, y_train,
    validation_split=0.2,
    epochs=100,
    batch_size=32,
    verbose=0
)

# Predict probabilities
y_prob3 = model3.predict(X_test).flatten()

# Convert to class labels
y_pred3 = (y_prob3 >= 0.5).astype(int)

# Evaluate
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, roc_auc_score

metrics3 = {
    "Accuracy": accuracy_score(y_test, y_pred3),
    "Precision": precision_score(y_test, y_pred3),
    "Recall": recall_score(y_test, y_pred3),
    "F1-Score": f1_score(y_test, y_pred3),
    "ROC-AUC": roc_auc_score(y_test, y_prob3)
}

metrics3["Model"] = "ANN - Simple SELU Nadam"
metrics3
```

Model 4 - Complex ANN with Dropout + ReLU + Adam

```
# Model 4 - Complex ANN with Dropout + ReLU + Adam (Classification)

model4 = tf.keras.Sequential([
    tf.keras.layers.Dense(128, activation='relu', input_shape=(X_train.shape[1],)),
    tf.keras.layers.Dropout(0.3),
    tf.keras.layers.Dense(64, activation='relu'),
    tf.keras.layers.Dropout(0.3),
    tf.keras.layers.Dense(32, activation='relu'),
    tf.keras.layers.Dense(1, activation='sigmoid') # classification output
])

# Compile
model4.compile(
    optimizer=tf.keras.optimizers.Adam(learning_rate=0.001),
    loss='binary_crossentropy',
    metrics=['accuracy']
)

# Train
history4 = model4.fit(
    X_train, y_train,
    validation_split=0.2,
    epochs=150,
    batch_size=32,
    verbose=0
)

# Predict probabilities
y_prob4 = model4.predict(X_test).flatten()

# Convert to class labels
y_pred4 = (y_prob4 >= 0.5).astype(int)

# Evaluate
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, roc_auc_score

metrics4 = {
    "Accuracy": accuracy_score(y_test, y_pred4),
    "Precision": precision_score(y_test, y_pred4),
    "Recall": recall_score(y_test, y_pred4),
    "F1-Score": f1_score(y_test, y_pred4),
    "ROC-AUC": roc_auc_score(y_test, y_prob4)
}

metrics4["Model"] = "ANN - Complex ReLU Dropout Adam"
metrics4
```

Model 5 - Large ANN with Dropout + SELU + RMSprop

```
# Model 5 - Large ANN with Dropout + SELU + RMSprop (Classification)

model5 = tf.keras.Sequential([
    tf.keras.layers.Dense(256, activation='selu', input_shape=(X_train.shape[1],)),
    tf.keras.layers.Dropout(0.4),
    tf.keras.layers.Dense(128, activation='selu'),
    tf.keras.layers.Dropout(0.4),
    tf.keras.layers.Dense(64, activation='selu'),
    tf.keras.layers.Dense(1, activation='sigmoid') # classification output
])

# Compile
model5.compile(
    optimizer=tf.keras.optimizers.RMSprop(learning_rate=0.001),
    loss='binary_crossentropy',
    metrics=['accuracy']
)

# Train
history5 = model5.fit(
    X_train, y_train,
    validation_split=0.2,
    epochs=200,
    batch_size=32,
    verbose=0
)

# Predict probabilities
y_prob5 = model5.predict(X_test).flatten()

# Convert to class labels
y_pred5 = (y_prob5 >= 0.5).astype(int)

# Evaluate
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, roc_auc_score

metrics5 = {
    "Accuracy": accuracy_score(y_test, y_pred5),
    "Precision": precision_score(y_test, y_pred5),
    "Recall": recall_score(y_test, y_pred5),
    "F1-Score": f1_score(y_test, y_pred5),
    "ROC-AUC": roc_auc_score(y_test, y_prob5)
}

metrics5["Model"] = "ANN - Large SELU Dropout RMSprop"
metrics5
```